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Cross-chain-based multi-microgrid energy trading mechanism

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HIGHLIGHTS

- A cross-chain-based multi-microgrid energy trading two-layer framework is developed.
- A multi-dimensional evaluation model of microgrids is proposed.
- A multi-microgrid energy trading mechanism based on multi-dimensional evaluation and reputation-based consensus is proposed.

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ABSTRACT

The increasing integration of distributed energy resources (DERs) and the emergence of multi-microgrid systems have necessitated the development of efficient and decentralized energy trading mechanisms. Traditional centralized energy trading frameworks face challenges related to scalability, security, market flexibility, and transaction bottlenecks, limiting their applicability in dynamic and distributed energy markets. Decentralized trading approaches, including peer-to-peer (P2P) energy trading, distributed optimization methods, and blockchain-based mechanisms, have been proposed to enhance energy exchange efficiency. However, existing blockchain-based energy trading models suffer from interoperability constraints, hindering seamless transactions between microgrids operating on different blockchain networks. To address these challenges, this paper proposes a cross-chain-based multi-microgrid energy trading mechanism that integrates blockchain interoperability protocols with decentralized trading frameworks. A two-layer cross-chain trading architecture is developed to enable secure, reliable, and seamless energy transactions across blockchain platforms. Furthermore, a multi-dimensional evaluation model (MEM) is introduced to comprehensively assess microgrid trading performance, incorporating key factors influencing energy transactions. Based on MEM, a multi-microgrid energy trading mechanism with reputation-based consensus is designed to optimize trading efficiency, mitigate energy imbalances, and ensure fairness in the market. Additionally, a dynamic price adjustment mechanism is developed, leveraging a multi-dimensional evaluation index (MEI) and supply-demand ratio (SDR) to enhance transaction stability and adaptability. The proposed cross-chain trading mechanism is validated through extensive simulations, demonstrating superior efficiency and enhanced system scalability compared to traditional energy trading mechanisms. The results confirm that the integration of cross-chain interoperability, multi-dimensional evaluation, and reputation-based consensus significantly improves market performance, energy utilization and social welfare. This study contributes to the advancement of scalable, secure, and fair multi-microgrid energy trading, offering a promising solution for future distributed energy markets.

1. Introduction

The increasing integration of distributed energy resources (DERs), including renewable energy sources such as photovoltaic (PV) and wind power, has significantly transformed the traditional power system. Conventional centralized energy trading mechanisms, which rely on a central authority for market clearing and transaction validation, face several challenges in this new energy landscape. The intermittency and unpredictability of renewable resources, along with the decentralized nature of microgrids (MGs), require innovative market structures to

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ensure efficient energy trading, optimal resource allocation, and system stability.

Traditional centralized electricity trading mechanisms optimize power dispatch based on global optimization models, which aim to achieve system-wide efficiency. However, these methods encounter significant limitations when applied to multi-microgrid energy trading, including scalability, security and privacy, market flexibility, and transaction bottlenecks. With the growing number of DERs and microgrids, centralized optimization methods become computationally intensive, making real-time market clearing impractical. The central authority has complete control over transaction data, increasing the risk of data breaches and single points of failure. Conventional systems struggle to accommodate P2P trading, which is essential for multi-microgrid energy systems to maximize local energy exchange. A single clearing entity may lead to transaction delays, reducing market efficiency and increasing operational costs. These challenges highlight the necessity for alternative market structures.

To address these challenges, decentralized trading mechanisms have emerged as promising solutions. Instead of relying on a central entity, these mechanisms enable local decision-making, allowing microgrids to trade electricity autonomously. Generally speaking, decentralized trading mechanisms include P2P energy trading, distributed optimization methods, and blockchain-based energy trading. The P2P energy trading enables direct energy transactions between microgrids without intermediaries. A variety of algorithms and solution strategies have been developed in the field of distributed optimization. Representative methods include the alternating direction method of multipliers (ADMM)[1][2], distributed gradient method[3], distributed Lagrangian method[4][5], and consistency-based iterative algorithm[6], which distribute computational loads across participants, improving scalability. The blockchain-based energy trading ensures transparency, security, and automated execution of smart contracts, eliminating the need for trusted intermediaries[7].

However, despite these advancements, existing blockchain-based energy trading frameworks face interoperability challenges when applied to multi-microgrid environments, where multiple blockchain networks may be used by different microgrid clusters. Current blockchain-based microgrid energy trading mechanisms suffer from interoperability constraints, limiting energy exchange across different networks. Cross-chain energy trading addresses these issues by enabling secure and efficient transactions between microgrids operating on different blockchain platforms.

Unlike traditional distribution networks, microgrids are typically operated by independent entities and possess self-contained control and management capabilities. However, due to the intermittency of renewables and the variability of local demand, a single microgrid often struggles to achieve internal energy balance at all times. Relying on trading with the utility grid may be economically suboptimal or constrained by grid regulations. Cross-microgrid energy trading provides a more flexible and cost-effective alternative by allowing surplus and deficit microgrids to exchange energy locally. However, since microgrids belong to different stakeholders and form a cross-trust-domain environment, establishing trust and ensuring secure cooperation becomes challenging. Our motivation lies in the unique need for interoperable and trustworthy trading mechanisms specifically tailored for microgrids, which exhibit (1) frequent local power imbalances, (2) reluctance to rely solely on centralized utilities, and (3) mutual distrust when operated independently. To address this, we adopt a cross-chain architecture that enables trustless interoperability among independently managed microgrid blockchains. This approach ensures transaction integrity, privacy protection, and consensus fairness in a decentralized setting, making it well-suited for the heterogeneity and autonomy of multi-microgrid systems.

Moreover, microgrids may operate in two distinct modes: grid-connected and islanded. In grid-connected mode, they can rely on the main grid for energy balancing and ancillary services. In contrast, during islanded operation—caused by faults, maintenance, or economic incentives—microgrids must independently manage their internal supply-demand balance. Our proposed cross-chain architecture is designed to support both modes. In the grid-connected mode, it enables secure, decentralized, and efficient P2P energy trading among multiple microgrids, serving as an alternative to centralized trading coordination and enhancing interoperability across heterogeneous microgrids. In the islanded mode, where centralized support is unavailable, the mechanism allows microgrids to autonomously trade surplus and deficit energy through a decentralized, trustless infrastructure built on cross-chain blockchain communication. This ensures continuous energy balance, increases system resilience, and enables microgrids to collaboratively manage energy in isolation from the main grid.

The cross-chain mechanism allows multiple microgrid clusters using different blockchain architectures to trade energy seamlessly. Cross-chain protocols maintain data integrity and transaction transparency, preventing fraudulent activities. Beyond these advantages, cross-chain energy trading offers additional benefits, including enhanced efficiency and security. This study proposes a cross-chain-based multi-microgrid energy trading mechanism, integrating blockchain interoperability protocols with decentralized trading frameworks. The key contributions of this work are summarized as follows:

1. To address interoperability and scalability challenges in decentralized multi-microgrid energy trading, we propose a novel cross-chain-based two-layer architecture. At the lower layer, each microgrid maintains its own blockchain to support autonomous and secure local energy transactions. At the upper layer, we design a dedicated relay chain that acts as a decentralized communication and coordination hub. Unlike conventional hub-and-spoke or oracle-based approaches, our relay chain enables trustless interoperability among heterogeneous microgrid blockchains by orchestrating cross-chain data exchange, bid aggregation, and reputation-weighted consensus through smart contracts. This design ensures data consistency, enhances transaction security, and supports scalable integration of grid-connected and islanded microgrids, laying the foundation for a practical and robust distributed energy trading infrastructure.
2. A multi-dimensional evaluation model (MEM) is proposed to assess trading-related characteristics of microgrids, enabling fairer and more efficient energy transactions, which comprehensively considers multiple key factors that affect the transaction results of multiple microgrids. In addition, based on this model, a multi-dimensional evaluation index (MEI) that dynamically reflects the current evaluation status of participating microgrids during the trading process is proposed to facilitate multi-microgrid energy trading.
3. A multi-microgrid energy trading mechanism based on MEM and reputation-weighted consensus is proposed. Each microgrid is assigned a dynamic reputation score, updated dynamically based on its historical trading behavior, protocol compliance, and market contribution. During the matching phase, the initiation order is determined by the current status of MEI, while current status of supply-demand conditions is also incorporated to enhance allocation fairness. In the consensus phase, transaction validation results are weighted by the reputation scores of participating microgrids. The final consensus is not based on a simple majority vote, but on the aggregate weighted reputation, ensuring that high-reputation microgrids exert greater influence on the decision outcome.

The remainder of this paper is organized as follows. The literature review is presented in Section 2. The multi-microgrid system framework for energy trading is presented in Section 3. Section 4 describes the P2P market mathematical models and optimization problem within a single microgrid. The multi-microgrid energy trading market and trading mechanism are presented in Section 5. Cross-chain system implementation for multi-microgrid energy trading is presented in Section 6. Section 7 discusses experimental validation and performance analysis. Section 8 concludes the paper and outlines future research directions.

2. Literature review

Traditional microgrid power trading mechanisms primarily rely on centralized optimal scheduling models, where a central operator makes global decisions to optimize economic dispatch. Nwulu et al.[8] proposed a mathematical optimization model for the economic scheduling of microgrids containing solar photovoltaic, diesel, and wind power generation resources. By incorporating an incentive-based demand response (DR) program, the study reduced fuel and trading costs while improving economic efficiency. Similarly, Santos et al.[9] integrated Energy Management System (EMS) strategies into the HOMER Pro software, optimizing day-ahead scheduling and energy trading between PV, battery storage, and the power grid. However, centralized approaches face significant challenges:

- Scalability Issues: High computational overhead when dealing with large-scale distributed energy systems.
- Privacy and Security Risks: Requires full data sharing with a central authority, increasing data exposure risks.
- Single Point of Failure: Vulnerable to system-wide disruptions due to reliance on a central controller.

These limitations have motivated the shift toward distributed and decentralized energy trading models.

To overcome the inefficiencies of centralized methods, researchers have explored distributed optimization algorithms for energy trading. These approaches decentralize decision-making, allowing individual microgrids to optimize their operations independently while still ensuring market efficiency. Several studies have applied primal-dual gradient-based optimization for decentralized electricity trading. Khorasany et al.[10] proposed a fully decentralized bilateral energy trading system, using primal-dual gradient methods to achieve market clearing without a central entity. Sorin et al.[11] introduced a relaxed consensus-based optimization framework to handle multi-bilateral transactions efficiently. However, primal-dual methods face slow convergence rates, requiring high communication overhead for frequent information exchange. To improve convergence efficiency, ADMM has been widely adopted for decentralized trading. Baroche et al.[12] and Guo et al.[13] implemented ADMM-based P2P trading models, optimizing social welfare and market fairness. References[14] proposed a novel online optimization framework that employs the Online Consensus Alternating Direction Method of Multipliers (OC-ADMM) algorithm to maximize social welfare. References[15] extended ADMM with asynchronous online consensus, enabling faster computation by allowing participants to proceed without waiting for all responses. While ADMM offers better scalability, existing ADMM-based models still lack cross-network interoperability, making inter-microgrid transactions inefficient.

As microgrids expand beyond isolated energy networks, research has shifted toward inter-microgrid trading mechanisms. Masaud et al.[16] introduced a smart contract-based two-layer architecture, enabling secure P2P trading between interconnected microgrids. Wei et al.[17][18] proposed robust optimization and Nash bargaining models, addressing fairness and uncertainty in inter-microgrid transactions. Lee et al.[19] formulated a Stackelberg game model, demonstrating how proportional energy and revenue allocation can maximize collective benefits. Jadhav et al.[20] incorporated aggregator-based game theory, preventing market manipulation by selfish microgrids. Despite these advances, existing inter-microgrid mechanisms still face interoperability challenges, as they do not support cross-platform energy exchange.

Blockchain technology has been introduced to enhance security, transparency, and automation in energy trading[21]. AlSkaif et al.[7] leveraged Hyperledger Fabric and ADMM to optimize blockchain-based P2P energy trading. Ping et al.[22] applied Shamir secret sharing and privacy-preserving-Byzantine-fault-tolerant (PP-BFT) algorithms for privacy-preserving transactions. Umar et al.[23] integrated blockchain with EMS, enabling smart contract-based transactions using cryptocurrency. However, most blockchain-based models are isolated within single networks, limiting interoperability across multiple microgrid markets.

Existing blockchain-based microgrid energy trading models remain largely confined to single blockchain platforms, posing significant barriers to interoperability. Although recent studies have explored cross-chain energy trading, major challenges still persist. Zhong et al.[24] introduced a cross-chain carbon-electricity trading model, but it focused only on intra-microgrid transactions. Zhang et al. [25] proposed a decentralized cross-chain energy trading model based on blockchain, heterogeneous energy blockchain transaction model based on the Energy Internet (EI-HEB), which aimed to solve the problem of efficient energy trading in heterogeneous energy networks in the Energy Internet, yet it lacked support for multi-microgrid. Wang et al. [26] proposed an electricity-carbon-TGC cross-chain market framework, but it did not address blockchain scalability issues. Thus, despite initial efforts in cross-chain trading, a comprehensive, scalable, and fully interoperable cross-chain microgrid energy trading framework is still lacking.

Recent studies have extended decentralized trading mechanisms beyond electricity-only systems to encompass multi-carrier energy networks, enabling more integrated and flexible energy management strategies. Daneshvar et al.[27] proposed a transactive energy framework specifically tailored for community microgrids within modern multi-carrier networks, which enables optimal scheduling and energy exchange among 100 % renewable-based microgrids. Their model highlights the potential of coordinated trading in enhancing techno-economic and environmental performance in centrally managed multi-carrier energy networks.

In parallel, community energy storage (CES) aggregation has emerged as a viable solution for improving energy autonomy in smart city contexts. Ghaddar et al.[28] introduced a CES-based clustering framework, where prosumers and consumers are grouped into energy-sharing communities. Their optimization model focuses on maximizing self-consumption and self-sufficiency while addressing network design constraints for efficient energy sharing.

These studies offer valuable insights for designing resilient and scalable architectures, and underscore the importance of integrating decentralized optimization, community-level coordination, and multi-energy flexibility—principles that inform the design of our proposed cross-chain trading mechanism.

The literature review reveals that microgrid energy trading has many challenges. Centralized optimization methods are inefficient for scalable distributed energy systems. Distributed optimization and ADMM-based trading mechanisms improve scalability, but lack cross-platform compatibility. Blockchain-based energy trading enhances security and automation, but remains confined to single-blockchain networks. Existing cross-chain trading models provide partial interoperability, yet fail to support dynamic multi-microgrid transactions. To address these challenges, this research proposes a cross-chain-based multi-microgrid energy trading mechanism, enabling seamless, secure, and efficient cross-microgrid energy exchange.

The comparison of existing related work with our work is shown in Table 1.

Compared with the existing literature summarized in Table 1, our proposed method demonstrates several distinct advantages in the context of microgrid energy trading. Our mechanism integrates a two-layer cross-chain framework with dynamic reputation-based consensus and multi-dimensional evaluation—features not jointly explored in prior works. Unlike many centralized or single-chain models with limited trading capabilities, our two-layer cross-chain framework enables

Table 1
Comparison of existing related work with our work.

Paper	Method	Decentralized?	Use optimization?	Use blockchain?	Trading scope	Scalable?	Interoperable across MGs?
[8]	Centralized optimization	No	Yes	No	No trading	Yes	No
[9]	EMS-based Mixed-Integer Linear Programming(MILP) integrated with HOMER Pro	No	Yes	No	No trading	Yes	No
[10]	Primal-Dual Gradient-based Bilateral Trading	Yes	Yes	No	P2P Market	Yes	No
[11]	Consensus-Based Approach with RCI	Yes	Yes	No	P2P Market	No	No
[12]	Exogenous cost allocation with network charges using ADMM	Yes	Yes	No	P2P Market	No	No
[13]	Chance-Constrained P2P Joint Energy and Reserve Market with Consensus ADMM	Yes	Yes	No	P2P Market	Yes	No
[14]	Online Consensus ADMM	Yes	Yes	No	P2P Market	Yes	No
[15]	Asynchronous Online Consensus ADMM	Yes	Yes	No	P2P Market	Yes	No
[16]	Two-layer architecture with smart contracts	Yes	Yes	Yes	Inter-MG	Yes	Yes
[17]	Robust optimization + Nash bargaining	Yes	Yes	No	Inter-MG	Yes	Yes
[18]	AUQ-ADMM	Yes	Yes	No	Inter-MG	Yes	Yes
[19]	Game-Theoretic Model (Stackelberg Game)	Yes	Yes	No	Intra-MG + Inter-MG	Yes	Yes
[20]	Aggregator-based game model + pricing mechanism	Yes	Yes	No	Inter-MG	Yes	Yes
[7]	Supply-demand matching and distance-based strategies + ADMM	Yes	Yes	Yes	P2P Market	Yes	No
[22]	PP-BFT algorithm + Shamir secret sharing	Yes	No	Yes	P2P Market	Yes	No
[23]	Blockchain-based P2P energy trading with battery storage	Yes	No	Yes	Intra-MG	Yes	No
[24]	Local Electricity and Carbon Trading with Cross-Chain Interaction	No	Yes	Yes	Intra-MG + Inter-MG	Yes	Yes
[25]	Cross-chain trading in EI-HEB	Yes	No	Yes	Intra-Sub-chains + Inter-Sub-chains	Yes	No
[26]	Electricity-carbon-TGC cross-chain framework	Yes	Yes	Yes	Intra-Market + Inter-Market	Yes	No
[27]	Hybrid DRCC and SP-based Transactive Energy Model	Yes	Yes	No	Intra-MG	Yes	Yes
[28]	MILP with Benders Decomposition	No	Yes	No	Intra-MG + Inter-MG	Yes	Yes
Our paper	**Cross-chain-based + Consensus ADMM + MEM + Reputation-based consensus**	**Yes**	**Yes**	**Yes**	**Intra-MG + Inter-MG**	**Yes**	**Yes**

both intra-microgrid and inter-microgrid collaborative energy trading. Within individual microgrids, decentralized execution and parallel processing of power transactions are achieved through the integration of ADMM and blockchain technologies. The use of Consensus ADMM combined with a smart-contract-enabled blockchain allows efficient decentralized optimization and parallel processing of transactions within microgrids. For inter-microgrid trading, trusted cross-chain data exchange is combined with a MEM and reputation-based consensus mechanism, ensuring secure, efficient, and fair energy transactions across heterogeneous microgrid systems. Scalability is further enhanced through the relay chain design, which aggregates and verifies cross-microgrid transactions. We introduce a MEI and a reputation-based consensus mechanism to dynamically assess trading behavior and mitigate manipulation, which are largely overlooked in existing models. The proposed framework is implemented as a prototype system and validated through simulations under multiple scenarios, showing promising results in terms of performance, resilience, and trustworthiness.

3. Multi-microgrid system framework for energy trading

In this study, we focus on a day-ahead energy trading market with a 24-h planning horizon, where market participants submit forecast-based bids, and trading decisions are finalized ahead of real-time operations. In this context, renewable energy generation and load demand are assumed to be known in advance through forecasting, and energy prices are negotiated directly among microgrids rather than being determined by a centralized market operator. Therefore, we do not explicitly model uncertainties such as renewable intermittency, load fluctuations, or market price volatility in the current framework. While the current study assumes deterministic inputs for tractability and clarity, the proposed architecture is designed to be extensible and can accommodate uncertainty-aware trading mechanisms in future work.

This section presents the overall architecture of the Multi-Microgrid System, as illustrated in Fig. 1. The proposed design adopts a two-layer architecture for microgrid energy trading, leveraging cross-chain technology. At the lower layer, various DERs within each microgrid are integrated with a blockchain platform to facilitate local transactions. The upper layer connects different microgrids to a relay chain platform via their respective blockchain networks, enabling secure and interoperable cross-chain communication.

We design a multi-microgrid system consisting of M interconnected microgrids, represented as $G = \{1, \dots, M\}$. Each microgrid incorporates various DERs, including conventional generators, wind power, residential users, and battery energy storage system (BESS).

Energy exchange between adjacent entities within a microgrid is enabled through bidirectional power links. Each microgrid's blockchain platform independently monitors and records P2P energy trading among

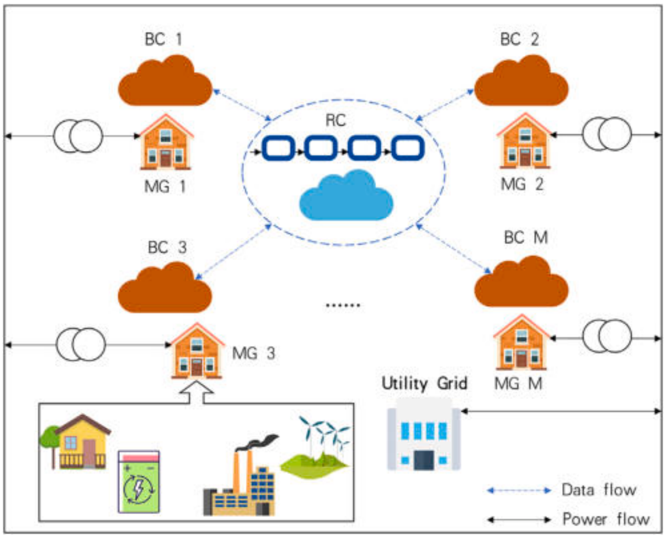

Fig. 1. Multi-microgrid system framework.

its entities. In cross-microgrid energy trading, microgrids function as either producers or consumers, depending on their energy surplus or deficit, allowing them to buy or sell energy as needed.

At the lower layer, participants within a microgrid negotiate energy transactions through P2P mechanisms to achieve local energy balance. A smart contract-based and Consensus ADMM method optimizes transaction allocation, ensuring accurate power exchange calculations for each microgrid. The local blockchain ledger records all P2P transactions, providing immutable and verifiable trading data, while smart meters validate power delivery to prevent fraudulent transactions.

At the upper layer, each microgrid blockchain connects to a dedicated relay chain, which serves as the core of our cross-chain system architecture. The relay chain functions as a decentralized coordination hub, enabling seamless bidirectional communication between heterogeneous microgrid blockchains. Unlike oracle-based or centralized gateway approaches, our relay chain ensures trustless interoperability by executing smart contracts and validating transactions across chains through a reputation-weighted consensus mechanism. Each microgrid registers on the relay chain with a unique identity, public key, and organizational credentials, after which energy bids, preference lists, and verified transaction results are transmitted and synchronized through the relay layer. This decentralized consensus framework eliminates reliance on a single entity, thereby avoiding single points of failure and reducing dependency on third-party intermediaries. The relay chain supports a large number of microgrids, enabling secure, auditable, and tamper-proof communication and transactions across heterogeneous systems. This flexibility makes it highly adaptable to dynamic energy trading markets compared to static direct connections or centralized platforms.

4. P2P energy trading within a single microgrid

This section presents the P2P energy trading market model and formulates the corresponding optimization problem within a single microgrid.

4.1. P2P energy trading modeling and optimization problem formulation

4.1.1. Mathematical model of participants

All participants in the microgrid engage in decentralized P2P energy trading, aiming to maximize their own interests. Each participant independently determines transaction prices and energy volumes.

A local microgrid market consists of n participants, represented by the set Ω, which includes producers, consumers, and prosumers. Specifically, Ω_g denotes the set of producers, Ω_c represents the consumers, and Ω_p refers to prosumers. Let ω_n be the set of neighboring participants of participant n. The net power injection of participant n, denoted as E_n, is the sum of the bilateral transactions between participant n and its trading counterparts. If $E_{nm} > 0$, participant n acts as a producer, selling energy to participant m. Conversely, if $E_{nm} < 0$, participant n is a consumer, purchasing energy from participant m. Here, m represents any participant in the set ω_n, which includes all participants except n. The transaction volume between participants n and m is denoted as E_{nm}, and the set of all bilateral transactions involving participant n is represented as $\mathbf{E_n} = \{E_{n1}, \dots, E_{nm}\}$.

The net power injection E_n of participant n is the total sum of its transaction volumes with all neighboring participants in ω_n, expressed as:

$$E_n = \sum_{m \in \omega_n} E_{nm}, \quad \forall n \in \Omega. \tag{1}$$

The power set-point of participant n is subject to the following constraint, ensuring it remains within the predefined power limits $\underline{E_n}$ and $\overline{E_n}$:

$$\underline{E_n} \le E_n \le \overline{E_n}, \quad \forall n \in \Omega. \tag{2}$$

In this study, producers primarily consist of conventional generators and wind power generators, while consumers include electric vehicles and residential users. The BESS is classified as a prosumer, capable of both consuming and supplying energy.

The power generation of both conventional and wind power generators is constrained within their respective upper and lower limits:

$$\underline{E_c} \le E_c \le \overline{E_c} \tag{3a}$$

$$\underline{E_w} \le E_w \le \overline{E_w} \tag{3b}$$

The power demand of household users is constrained within their respective upper and lower limits:

$$\underline{E_u} \le E_u \le \overline{E_u} \tag{4}$$

As a prosumer, the BESS has a net power output that equals the difference between its charging and discharging power:

$$E_b = E_{bc} - E_{bd} \tag{5}$$

The State of Charge (SoC) of the BESS at time t is determined by the following equation:

$$SoC_b^t = SoC_b^{t-1} + (\eta_c E_{bc}^t - \frac{E_{bd}^t}{\eta_d})\frac{\Delta_t}{Q_B} \tag{6}$$

where η_c and η_d represent the charging and discharging efficiencies, respectively, and Q_B denotes the battery capacity.

The operation of the BESS is further constrained by the following limits on its SoC and charging/discharging power:

$$\underline{SoC_b} \le SoC_b \le \overline{SoC_b} \tag{7a}$$

$$0 \le E_{bd} \le \omega_{dch} \overline{E_{bd}} \tag{7b}$$

$$0 \le E_{bc} \le \omega_{ch} \overline{E_{bc}} \tag{7c}$$

where ω_{ch} and ω_{dch} are binary variables ensuring that charging and discharging do not occur simultaneously, satisfying the complementary constraint:

$$\omega_{ch} + \omega_{dch} \le 1. \tag{8}$$

4.1.2. Power network model

The power system consists of N nodes. To account for physical network constraints, a System Operation Node (SON) is introduced into the P2P market. The SON acts as a line manager, responsible for calculating power flow without interfering with the direct P2P transactions between participants. This ensures that the decentralized transaction structure remains intact while maintaining network feasibility.

The internal power flow within the microgrid follows the AC power flow model. The SON formulates an optimization problem based on second-order cone relaxation, utilizing the DistFlow model to compute power flows and bus voltage magnitudes[29][30].

For each node, the total input power must balance with the sum of output power, local loads, and transmission losses. The power flow equations at the nodes, derived using second-order cone relaxation, are given as follows:

$$P_j = \sum_{j\to k} P_{jk} - \sum_{i\to j}(P_{ij} - R_{ij}L_{ij}), \quad \forall i,j,k \in N \tag{9}$$

$$Q_j = \sum_{j\to k} Q_{jk} - \sum_{i\to j}(Q_{ij} - X_{ij}L_{ij}), \quad \forall i,j,k \in N \tag{10}$$

The voltage balance equation is expressed as follows:

$$\begin{aligned} U_j = U_i &- 2*(R_{ij}P_{ij} + X_{ij}Q_{ij}) \\ &+ (R_{ij}^2 + X_{ij}^2)L_{ij}, \quad \forall i,j \in N \end{aligned} \tag{11}$$

The second-order cone constraint is formulated as:

$$\left\| \begin{matrix} 2P_{ij}^2 \\ 2Q_{ij}^2 \\ L_{ij} - U_i \end{matrix} \right\|_2 \le L_{ij} + U_i, \quad \forall i,j \in N \tag{12}$$

The constraints on generator active and reactive power are defined as follows:

$$\underline{P_i} \le P_i \le \overline{P_i}, \quad \forall i \in N \tag{13a}$$

$$\underline{Q_i} \le Q_i \le \overline{Q_i}, \quad \forall i \in N \tag{13b}$$

The constraints on node voltage amplitude and line current amplitude are given by:

$$\underline{U_i} \le U_i \le \overline{U_i}, \quad \forall i \in N \tag{14a}$$

$$L_{ij} \le \overline{L}_{ij}, \quad \forall (i,j) \in \pounds \tag{14b}$$

where $U_i = (V_i)^2$ represents the squared voltage amplitude at node i, $L_{ij} = (I_{ij})^2$ denotes the squared current amplitude of branch (i,j), and $\pounds$ represents the set of all transmission lines. Additionally, R_{ij} and X_{ij} denote the resistance and reactance of the branch, respectively.

To maintain power balance at each bus of the microgrid, the following constraint must be satisfied:

$$\sum_{n\in N_i} E_n = \sum_{(i,j)\in\pounds} P_{ij}, \quad \forall i \in N \tag{15}$$

4.1.3. P2P trading product differentiation

P2P transactions within a single microgrid incorporate product differentiation. The bilateral trade costs are modeled as a linear function of the transaction volume between each participant and its adjacent participants:

$$\tilde{C_n}(\mathbf{E_n}) = \sum_{m\in\omega_n} c_{nm}E_{nm} \tag{16}$$

where c_{nm} represents the bilateral trade coefficient, which captures the differentiation of traded energy products. This coefficient is influenced by various factors, including carbon emissions, transmission distance, production and consumption scales, and other relevant parameters.

4.1.4. Cost and utility function for sellers and buyers

For modeling convenience, the seller's cost function and the buyer's utility function are both represented as quadratic convex functions [31] of active power:

$$C_n(E_n) = \frac{1}{2}\alpha_n E_n^2 + \beta_n E_n + \gamma_n \tag{17}$$

When a participant acts as a seller, the parameters α_n, β_n, and γ_n denote predetermined positive constants that define the seller's cost function. These parameters reflect the producer's willingness to sell energy at varying prices and are influenced by factors such as generation type, load conditions, and future risk considerations. When a participant acts as a buyer, α_n, β_n, and γ_n instead represent the buyer's utility function parameters, which characterize the buyer's responsiveness to price variations. Since each participant operates independently within the market, these parameters are unique to each entity and may vary not only between different participants but also across different time periods throughout the day.

4.1.5. Optimization problem formulation

The P2P energy trading mechanism within a microgrid aims to minimize the total cost, which is equivalent to maximizing social welfare. The objective function is to minimize the overall cost incurred by all participants. Taking into account the previously defined constraints, the optimization problem for P2P energy trading within a single microgrid is formulated as follows:

$$\min \sum_{n\in\Omega} C_n(E_n) + \tilde{C_n}(\mathbf{E_n}) \tag{18}$$

s.t. (1)-(15)

4.2. Decentralized negotiation mechanism within a single microgrid

The ADMM is an optimization algorithm that combines the decomposability of dual ascent with the convergence properties of the multiplier method. However, traditional ADMM implementations require a central coordinator to update dual variables. To adapt ADMM for decentralized P2P energy trading in a single microgrid, we employ the Consensus ADMM approach [13]. In this framework, each participant independently solves its own subproblem and iteratively updates its solution until convergence is achieved. Additionally, a SON is introduced as a single agent responsible for power flow calculations, ensuring that physical network constraints are met without interfering with P2P transactions.

For each node, the sum of input power must equal the total output power, including load and line losses. The physical network constraints (9)-(15) can be reformulated as the following optimization problem:

$$\mathbf{P^{k+1}} = \arg\min_{\mathbf{P}} \sum_{i\in N} \left\| \sum_{n\in N_i} E_n^k - \sum_{(i,j)\in\pounds} (P_{ij} + R_{ji}L_{ji}) \right\|_2^2 \tag{19}$$

s.t. (11)-(14b)

At each iteration of the negotiation mechanism, the SON first collects the sum of injected powers P_n^k from all agents connected to different buses. It then computes the updated power flow $\mathbf{P^{k+1}}$ and transmits the results back to all agents. To facilitate decentralized decision-making, dual variables are introduced to regulate power transactions. A Consensus ADMM-based distributed optimization algorithm is implemented to determine the optimal allocation of power resources across the microgrid. The local optimization subproblem for each agent n at bus i in iteration k is formulated as follows:

$$\begin{aligned}\mathbf{E_n^{k+1}} = \arg\min_{\mathbf{E_n}} C_n(E_n) + \tilde{C}_n + (\mathbf{E_n}) \\ \times \sum_{m\in\omega_n}\left[\lambda_{nm}^k\left(\frac{E_{nm}^k - E_{mn}^k}{2} - E_{nm}\right)\right. \\ \left. + \frac{\rho}{2}\left(\frac{E_{nm}^k - E_{mn}^k}{2} - E_{nm}\right)^2\right] \\ + v_i^k\left[E_n + E_{-n}^k - \sum_{(i,j)\in\pounds}\left(P_{ij}^{k+1} + R_{ji} * L_{ji}\right)\right] \\ + \frac{\tau}{2}\left[E_n + E_{-n}^k - \sum_{(i,j)\in\pounds}\left(P_{ij}^{k+1} + R_{ji} * L_{ji}\right)\right]^2\end{aligned} \tag{20}$$

Where P_{-n}^k represents the total power injected by all agents except n at bus i; λ_{nm} is a dual variable representing both the power trading price and the line price v_i. The power trading price λ_{nm}^{k+1} and the line price v_i^{k+1} are updated according to the following rules:

$$\lambda_{nm}^{k+1} = \left[\lambda_{nm}^k - \rho\left(\frac{E_{nm}^k - E_{mn}^k}{2}\right)\right]^+ \tag{21}$$

$$v_i^{k+1} = \left[v_i^k - \tau\left(\sum_{n\in N_i} E_n^{k+1} - \sum_{(i,j)\in\pounds}\left(P_{ij}^{k+1} + R_{ji}L_{ji}\right)\right)\right]^+ \tag{22}$$

Where $[\cdot]^+$ denotes $\max(0,\cdot)$, ensuring non-negativity. Given that the objective function is convex, the ADMM-based optimization method is guaranteed to converge to the global optimal solution. The market is considered converged when the total local residual and the SON objective function meet the global stopping criteria:

$$PR^{k+1} = \sum_{b\in N} r_n^{k+1} \le \chi^p \tag{23}$$

$$DR^{k+1} = \sum_{b\in N} s_n^{k+1} \le \chi^d \tag{24}$$

$$SON^{k+1} = \sum_{i\in N}\left\|\sum_{n\in N_i} E_n^k - \sum_{(i,j)\in\pounds}(P_{ij} + R_{ji}L_{ji})\right\|_2^2 \tag{25a}$$

$$SON^{k+1} \le \chi^{son} \tag{25b}$$

In this study, we consider scenarios where extreme conditions—such as excessive renewable energy generation or sudden surges in electricity consumption—prevent certain participants from achieving energy trading balance. To address this, we utilize primal and dual residuals as convergence criteria, formulated as follows:

$$r_n^{k+1} = \sum_{m\in\omega_n}(E_{nm}^{k+1} + E_{mn}^{k+1})^2 \tag{26}$$

$$s_n^{k+1} = \sum_{m\in\omega_n}(E_{nm}^{k+1} - E_{mn}^{k+1})^2 \tag{27}$$

If both the primal and dual residuals fall below their respective predefined thresholds, the market is considered balanced, and convergence is deemed achieved. However, if the dual residual is below the threshold while the primal residual remains above it—and there is no significant reduction between consecutive iterations—we determine that some participants cannot reach trading equilibrium. In such cases, the algorithm stops updating and initiates cross-microgrid energy trading. The following condition is used to identify a failure to converge and trigger early termination of the algorithm:

$$PR^k > \chi^p \tag{28a}$$

$$DR^k \le \chi^d \tag{28b}$$

$$|PR^k - PR^{k-1}| \le \chi^{\Delta p} \tag{28c}$$

4.3. Handling uncertainty within a single microgrid trading

Although our proposed inter-microgrid trading mechanism primarily focuses on day-ahead decisions based on forecasts, the issue of uncertainty is not overlooked. In fact, existing methods for modeling uncertainty within single-microgrid P2P trading offer valuable theoretical foundations and practical tools that can be extended and integrated into our cross-microgrid framework. Below, we briefly summarize the representative methods and highlight their compatibility with our design.

Stochastic Programming (SP) approaches typically incorporate forecast uncertainty by modeling renewable energy generation as probabilistic variables. For instance, our prior work[13] demonstrates how joint energy and reserve trading can be formulated using chance-constrained programming, enabling each agent to allocate appropriate reserves based on acceptable risk levels. This provides a principled way to internalize generation uncertainty while preserving decentralized decision-making. Given that our trading framework supports reserve coordination and agent-level optimization, such SP methods can be naturally integrated.

Robust Optimization (RO) approaches provide an alternative approach that avoids requiring precise probability distributions. Recent studies[32] have introduced fully data-driven Distributionally Robust Optimization (DRO) models, leveraging techniques like Deep Gaussian Processes (DGPs) and ensemble learning (e.g., Bagging) to construct ambiguity sets for uncertain variables such as PV output. These approaches optimize decisions under worst-case scenarios drawn from plausible distributions and are solved using decentralized methods like ADMM—consistent with our market-clearing architecture. This confirms that DRO-based techniques are structurally compatible with our mechanism and can be applied to enhance robustness against renewable intermittency.

Moreover, Mansouri et al. [33] present a hybrid robust-stochastic approach that combines Monte Carlo-based scenario generation with worst-case optimization. Their use of adaptive ADMM aligns with our coordination framework, suggesting that hybrid extensions are also readily compatible with our architecture.

In conclusion, the uncertainty modeling methods developed for single-microgrid P2P trading—whether stochastic, robust, or hybrid—can be effectively adapted to our inter-microgrid trading mechanism. Their integration strengthens the resilience of energy trading against forecast errors, generation variability, and demand uncertainty, without compromising the decentralized, flexible nature of our proposed design.

5. P2P energy trading among multiple microgrids

In Section 4, the consensus ADMM algorithm was employed for P2P energy trading within a single microgrid. This section extends the discussion to multi-microgrid energy trading, first by modeling the multi-microgrid trading market and then by proposing a decentralized energy trading mechanism based on cross-chain technology. This mechanism integrates an MEI for microgrids while ensuring transaction consensus through reputation-based validation among microgrids.

5.1. Multi-microgrid energy trading market modeling

A single microgrid may struggle to handle extreme conditions such as:

- Excessive renewable energy generation leading to surplus energy.
- Surging residential electricity consumption creating unexpected demand spikes.
- Generator failures causing supply shortages.

Under such circumstances, a single microgrid may fail to achieve internal energy balance, and the iterative negotiation method discussed in the previous section could result in many unresolved bilateral transactions. The accumulation of these transactions reflects the net energy deficit or surplus of the microgrid.

To overcome these limitations, a multi-microgrid energy trading framework is introduced. This system facilitates energy circulation and balance on a broader scale by interconnecting multiple microgrids. Each microgrid in this trading network operates as a prosumer—it can act as an energy supplier when it has surplus electricity or as an energy consumer when facing a deficit. The utility grid also participates in the market, serving as a backup energy source and ensuring system stability when microgrid-to-microgrid transactions cannot fully meet demand.

In a multi-microgrid energy trading scenario, the market consists of G participating microgrids, categorized into G_S sellers and G_B buyers, denoted as:

$$G_S = \{1, \dots, G_S\}, \quad G_B = \{1, \dots, G_B\} \tag{29}$$

where $G_S \cup G_B = G$ and $G_S \cap G_B = \oslash$, ensuring that no microgrid is both a buyer and a seller simultaneously. Each microgrid collects K pairs of bidding values $\{q_l, p_l\}$ for unresolved transactions, which are prioritized based on the absolute value product of each pair, $|p_l * q_l|$, where p_l and q_l represent the transaction price and volume difference of the l_{th} transaction of the microgrid, respectively. The net power balance of each microgrid is given by:

$$q_{net} = \sum_{l=1}^{K} q_l \tag{30}$$

where $q_{net} > 0$ means the microgrid has a surplus and acts as a seller and $q_{net} < 0$ means the microgrid has a deficit and acts as a buyer. The initial matching price for each microgrid p_{init} is computed as:

$$p_{init} = \sum_{l=1}^{K} p_l \frac{q_l}{q_{net}} \tag{31}$$

For each microgrid e, the energy balance constraint is formulated as:

$$\sum_{f \in G} E_{ef} + EU_e = \sum_{l=1}^{K_e} q_l \tag{32}$$

where E_{ef} represents the energy traded between microgrid e and f. and EU_e represents energy purchased from or sold to the utility grid.

The utility function of a seller microgrid e is:

$$U_e = \sum_{f \in G} (p_{ef} E_{ef}) + p_s^u EU_e \tag{33}$$

The cost function of a buyer microgrid e is:

$$C_e = \sum_{f \in G} (p_{ef} E_{ef}) + p_b^u EU_e \tag{34}$$

where E_{ef} and p_{ef} represent energy trading volume and clearing price between microgrid e and f, respectively. EU_e is the amount of energy purchased from or sold to the utility grid of microgrid e. If e is a producer, $E_{ef} > 0$, $EU_e \geq 0$; if e is a consumer, $E_{ef} < 0$, $EU_e \leq 0$. $p_b^u(E)$ and $p_s^u(E)$ represent the price functions of energy that microgrid buys from or sells to the utility grid.

To encourage market participation, the utility grid offers electricity price discounts to microgrids that engage in trading. The price functions are:

$$p_b^u(E) = \begin{cases} 0.95 * PBU, & |E| \leq 1 \\ 0.98 * PBU, & 1 < |E| \leq 2 \\ PBU, & 2 < |E| \end{cases} \tag{35a}$$

$$p_s^u(E) = \begin{cases} 1.05 * PSU, & E \leq 1 \\ 1.03 * PSU, & 1 < E \leq 2 \\ PSU, & 2 < E \end{cases} \tag{35b}$$

where PBU and PSU are the base buying/selling prices of the utility grid. The clearing price is constrained within:

$$PSU < p_{ef} < PBU \tag{36}$$

The welfare of sellers (denoted as WS) is linked to the producer surplus, which is the difference between actual revenue and the minimum acceptable price (PSU):

$$WS_s = U_s - PSU \sum_{l=1}^{K_e} q_l \tag{37}$$

The welfare of buyers (denoted as WB) is linked to consumer surplus, which is the difference between the maximum willingness to pay (PBU) and the actual cost paid:

$$WB_b = C_b - PBU \sum_{l=1}^{K_e} q_l \tag{38}$$

The whole multi-microgrid energy trading market social welfare refers to the total welfare of all participants in multi-microgrid energy trading. The objective function in this market which aims to maximize welfare of all participants can be expressed as follows:

$$\max \left(\sum_{b \in G_B} WB_b + \sum_{s \in G_S} WS_s \right) \tag{39}$$

s.t. (32), (36)

5.2. Multi-microgrid energy trading matching mechanism

This section introduces the MEM and the corresponding MEI for microgrids, followed by the reputation index and its dynamic adjustment mechanism. Furthermore, we integrate a transaction matching mechanism based on MEI and SDR to develop a reputation-based transaction consensus mechanism for cross-microgrid energy trading. By incorporating distance-based pricing, a dynamically adjusted bidirectional stable matching algorithm, and a reputation-based consensus mechanism, the proposed approach enhances market fairness, efficiency, and stability. This novel integration of methodologies ensures equitable transaction matching, mitigates dishonest behavior, and optimizes market dynamics, making it a promising solution for future decentralized energy markets.

5.2.1. Multi-dimensional evaluation model

The energy trading of multi-microgrids has problems of low efficiency, lack of fairness, and uneven resource allocation. To facilitate more fair and economical energy trading, a MEM is designed. This model takes into account several key factors that influence energy trading outcomes, such as the environmental impact, the distance between microgrids, the rate of successful energy trades, and the reasonableness of bidding prices. Each factor affects the transaction to varying degrees, but all contribute to the overall transaction results. The MEM incorporates multiple key factors that influence trading outcomes, including:

- Distance between microgrids(D) – Accounts for transmission losses and efficiency.

- Environmental impact(I_E) – Encourages the use of renewable energy sources.
- Transaction success rate(I_M) – Reflects a microgrid's historical ability to successfully complete trades.
- Bidding price rationality(I_B) – Evaluates whether a microgrid's bid aligns with market conditions.

Each of these factors plays a role in determining a microgrid's overall trading performance and market participation priority. The evaluation model dynamically adjusts transaction priorities, ensuring that well-performing microgrids receive appropriate incentives while discouraging inefficient or unfair trading practices.

The distance between microgrids significantly impacts both transaction costs and efficiency in energy trading. Unlike intra-microgrid trading, where energy losses are minimal, cross-microgrid energy trading considers line losses, which primarily depend on the physical separation between microgrids. Geographic proximity provides a competitive advantage—shorter distances reduce transmission costs and improve trading efficiency. Thus, distance is a key factor in determining the cost-effectiveness of multi-microgrid energy trading. The physical distance between two microgrids, denoted as $D_{phy,ef}$, represents the separation between microgrid e and microgrid f. To ensure dimensional consistency in evaluation, the distance is standardized into a distance evaluation score as follows:

$$D_{ef} = 1 - \frac{D_{phy,ef} - D_{phy,\min}}{D_{phy,\max} - D_{phy,\min}} \tag{40}$$

where $D_{phy,\max}$ and $D_{phy,\min}$ denote the maximum and minimum distances among all microgrids, respectively. This transformation ensures that closer microgrids receive a higher score (closer to 1), while distant microgrids receive a lower score (closer to 0). This normalized metric facilitates fair and consistent evaluation across different trading scenarios.

To encourage renewable energy use and reduce emissions, we integrate environmental impact factors into energy trading. These factors prioritize and price transactions based on a microgrid's renewable energy share: higher renewable usage yields a better score, incentivizing greener energy adoption. This promotes sustainability, market efficiency, and lower carbon emissions. The impact index is calculated from the renewable proportion in a microgrid's total energy as follows:

$$I_E = \frac{E_g}{E_t} \tag{41}$$

where E_g represents the renewable energy used by the microgrid, and E_t denotes its total energy consumption.

The successful trading rate index quantifies a microgrid's transaction success rate, reflecting its ability to execute trades efficiently. This index is determined by the volume of successful trade matches within a given trading round. A higher success rate suggests that a microgrid's bids are reasonable, responses are timely, and transactions are stable, thereby improving its index score. The transaction success rate index is calculated as:

$$I_M = \frac{M_s}{M_t} \tag{42}$$

where M_s denotes the total successful energy traded, and M_t represents the total energy expected to be traded in that round.

The bid rationality index evaluates how closely a microgrid's bid (p_g) aligns with the market average price (p_m). A reasonable bid increases the likelihood of transaction success, as it improves the chances of being selected in the matching process. To quantify this, the bid rationality index is defined based on the relative deviation between the microgrid's bid and the market average price:

$$I_B = \left(1 - \frac{\left|p_g - p_m\right|}{p_m}\right) \tag{43}$$

A lower deviation results in a higher index value, indicating that the bid is more reasonable and better aligned with market expectations, thereby increasing the microgrid's chances of successful matching.

Finally, to effectively evaluate the current performance of microgrids in the transaction matching process, we propose an MEI based on the previously defined evaluation model. The MEI is formulated as:

$$MEI = \alpha_I I_E + \beta_I I_M + \gamma_I I_B \tag{44}$$

$$\alpha_I + \beta_I + \gamma_I = 1 \tag{45}$$

where α_I, β_I, γ_I are weight coefficients, determining the contribution of each factor to the overall evaluation.

The MEI is dynamically calculated based on the current evaluation status of participating microgrids, allowing the system to prioritize microgrids based on their performance in the energy trading market. A higher MEI indicates a microgrid with better trading behavior and higher reliability, which enhances its priority in transaction matching and partner selection.

Different from the price-based matching, in the matching process, the microgrids initiating matching are sorted according to their MEI scores. This generates an proposal matching priority list Γ, where higher-MEI microgrids are prioritized for trading.

Inspired by the work in [34], we further introduce a distance price adjustment based on the distance evaluation score. This mechanism considers geographical proximity, wherein closer microgrids experience lower transaction costs and higher matching priority. The distance price adjustment is detailed in Section 5.2.3.

5.2.2. Dynamic reputation evaluation mechanism

The MEI serves as a current assessment tool for microgrids' trading behavior and market performance, while reputation reflects their long-term credibility based on historical transactions. Reputation is essentially the cumulative result of multi-dimensional evaluations over time, with higher-reputation microgrids enjoying greater market trust and transaction facilitation.

The reputation index is dynamically updated after each transaction: it increases for successful executions, decreases for defaults, and remains unchanged for unmatched transactions. This adaptive mechanism incentivizes sustained high-quality trading, as microgrids with stronger reputations gain greater market trust, promoting consensus and system stability. In addition, considering that malicious nodes deliberately delay time and postpone verification, we introduce a delayed verification penalty term to take the consensus verification time into account in the dynamic reputation mechanism. This user who deliberately delays time will lose the consensus authority for this round of transactions and be punished. In more serious cases, they will permanently lose the consensus verification authority. The reputation index can be formulated as follows:

$$R = \begin{cases} R^{pre} + \sum_{k=1}^{r} \varphi_R(k)\omega_M^k + \alpha_R(1-\omega_b) - \beta_R\omega_b, P_{DV} = 0 \\ R^{pre} + P_{DV}, P_{DV} < 0 \end{cases} \tag{46}$$

where ω_b is a binary variable that takes a value of 0 or 1, depending on whether the microgrid has exhibited dishonest behavior. A value of 0 indicates no dishonest behavior, while a value of 1 signifies the occurrence of such behavior. Similarly, ω_M^k is a binary variable indicating whether the microgrid successfully matched in the k-th round. R^{pre} represents the reputation index from the previous evaluation cycle. The parameters α_R

and β_R denote the reward and penalty constants, respectively, where β_R penalizes dishonest behavior by reducing the reputation score, while α_R incentivizes microgrids to fulfill their obligations. P_{DV} is a penalty term for delayed verification. The reward function $\varphi_R(k)$, which quantifies the incentive associated with a successful match in the k-th round, is expressed as follows:

$$\varphi_R(k) = \begin{cases} e^{2-k}, k \le 3 \\ e^{-k}, k > 3 \end{cases} \tag{47}$$

This function emphasizes that the reputation incentives for successful matching are relatively high in the first three rounds, encouraging participants to make reasonable bids and reach agreements as early as possible. Beyond the fourth round, the rewards become negligible.

The delayed verification penalty function $P_{DV}(\tau_m)$ is related to the consensus verification time:

$$P_{DV}(\tau_m) = \begin{cases} -\delta_{DV}, \tau_m > \tau_{reasonable} \\ 0, otherwise \end{cases} \tag{48}$$

where τ_m represents the delayed verification time of node m in consensus. δ_{DV} is the delayed verification penalty factor. $\tau_{reasonable}$ is a reasonable delayed verification time threshold. When the node m's reputation is less than 0 ($R_m < 0$), the node loses consensus authority.

To prevent high-reputation microgrids from maintaining long-term dominance, a reputation decay mechanism is introduced. This ensures that historical advantages gradually diminish, promoting fair competition and preventing market monopolization. The reputation decay function is given by:

$$R = R^{pre} e^{-\delta t} \tag{49}$$

where δ represents the decay coefficient, determining the rate at which reputation decreases. Furthermore, by periodically re-evaluating and refining the evaluation metrics, the system can maintain fairness and ensure equitable opportunities for all participating microgrids.

5.2.3. Consensus mechanism based on reputation

In the previous section, we introduced the MEI model and the reputation index. This section presents a transaction consensus mechanism that leverages reputation to enhance fairness and efficiency in cross-microgrid energy trading. The reputation-based consensus mechanism facilitates cross-microgrid matching and transaction decision-making by incorporating reputation indices, thereby promoting more credible and efficient transactions.

(1) Distance-based pricing mechanism

To quantify the cost implications of geographical separation, the distance price between microgrid e and f is calculated as follows:

$$p^d_{s,ef} = \frac{p_{s,e}}{D_{ef}^{\frac{1}{2}}}, e \in G_S, f \in G_B \tag{50a}$$

$$p^d_{b,ef} = p_{b,e} D_{ef}^{\frac{1}{2}}, e \in G_B, f \in G_S \tag{50b}$$

where $p_{s,e}$ and $p^d_{s,ef}$ denote the initial price and distance-adjusted price for a selling microgrid e, respectively, while $p_{b,e}$ and $p^d_{b,ef}$ represent the corresponding values for a buying microgrid e.

Each microgrid computes its distance-adjusted price based on the geographical distance between itself and potential trading counterparts, forming a prioritized list of potential transaction partners. The priority ranking follows the basic trading condition, where a transaction is feasible if the buyer's bid meets or exceeds the seller's price (or vice versa). This results in an available matching preference list, denoted as Θ_{avail}.

Note that while distance-based pricing offers a scalable and intuitive metric for localizing trading incentives, its effectiveness may vary in sparse or congested microgrid topologies. Future work will consider integrating power flow constraints and dynamic congestion pricing to enhance location sensitivity and ensure fairness across heterogeneous networks.

(2) Dynamically adjusted bidirectional stable matching algorithm

This section introduces a modified Gale-Shapley algorithm for multi-microgrid energy trading. The standard Gale-Shapley algorithm, which typically prioritizes buyers as initiators, may lead to systemic biases. To enhance fairness, we introduce modifications to balance the priorities of both buyers and sellers. The fairness of the market is ensured by determining the market type based on supply and demand. This mechanism helps maintain an equitable trading environment for all participants.

Then, we present the matching process of this mechanism as shown in Fig. 2.

Market Type Determination: The market type—buyer's market or seller's market—is determined by the supply-demand balance. If supply exceeds demand, buyers initiate the matching process. Conversely, if demand surpasses supply, sellers take the lead in proposing transactions. This ensures a fair trading environment where the dominant side does not unilaterally dictate the market.

Matching Priority: Once the market type is established, a proposal priority list Γ is generated based on the MEI, indicating the order in which microgrids initiate matching. Microgrids with higher MEI values are granted higher priority when proposing transactions. Counterparties are invited based on their available matching preference list Θ_{avail}, which ranks potential partners by distance price—favoring geographically closer microgrids. This is different from the price-based priority mechanism and fully considers the multi-dimensional evaluation factors of microgrids.

Matching Process: The matching process proceeds as follows:

1. Buyers make transaction proposals: Each buyer submits a transaction proposal to sellers ranked in its available matching preference list Θ_{avail}.
2. Sellers process matching proposals: If a seller has no existing match, it accepts the first proposal and is temporarily matched. If a seller already has a match, it compares the new proposal with the existing one. If the new buyer is preferable, the seller switches matches, and the previously matched buyer must submit a new proposal to the next available seller at the end of the round.
3. Unmatched buyers continue to try: Buyers who have not successfully matched continue to send matching requests to the next seller, and matched sellers can also continue to adjust matching objects until all buyers and sellers reach a stable match, or all buyers have no more sellers to choose from.
4. Round termination condition: When all buyers no longer make matching requests, this round of matching ends and the matching process enters a stable state.
5. Post-matching adjustment: At this point, the system adjusts the prices and net power of buyers and sellers, updates their MEI, updates the market type, and enters the next round of matching.
6. Matching termination condition: This iterative process continues until all buyers and sellers reach a stable match or until no further matching is possible.

A similar process is followed in a seller's market, where sellers initiate transactions. Once a match is established, the final transaction price is set as the average of the seller's and buyer's bids. After each round, trading parameters—including prices, net power, and MEI values—are updated, and the next round begins. The algorithm terminates when only unmatched buyers or sellers remain in the market.

(3) Reputation-based consensus mechanism

Traditional consensus mechanisms, which typically rely on majority voting or fixed rules, often fail to differentiate between reputable and unreliable microgrids. As a result, low-reputation participants may exploit the system, distorting market fairness. In cross-microgrid energy

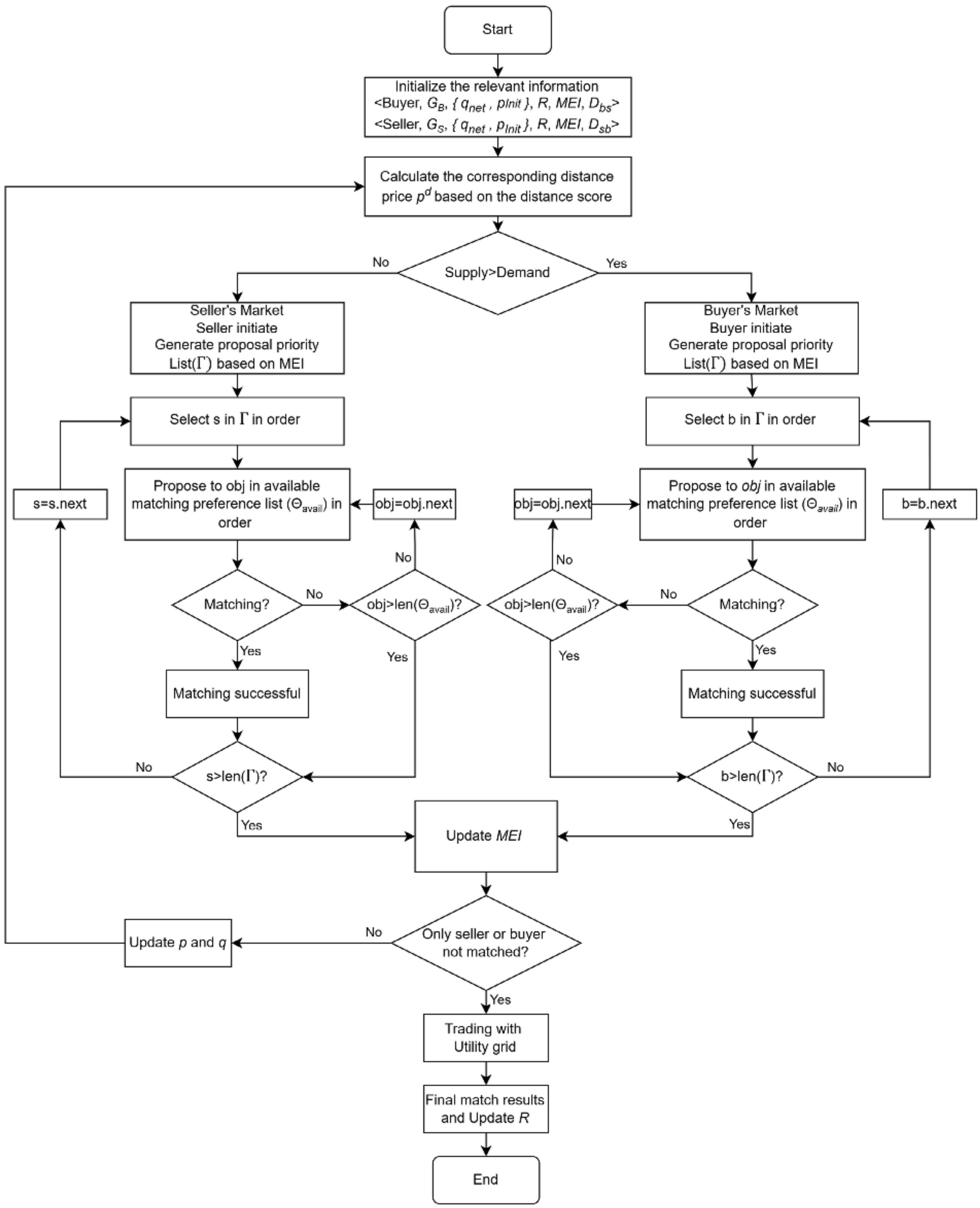

Fig. 2. Flowchart of matching process.

trading, such mechanisms may allow dishonest or inefficient microgrids to disrupt transactions, undermining market integrity. To address these challenges, we propose a reputation-based consensus mechanism that ensures transactions are validated in a fair and transparent manner.

The core of this mechanism is reputation-weighted transaction approval. Each microgrid is assigned a dynamic reputation score, updated after each transaction is completed. In each consensus round, participants are divided into agreeing (G_a) and disagreeing (G_d) groups. A transaction is approved if the total reputation of agreeing microgrids outweighs that of the opposing group:

$$\sum_{a\in G_a} R_a - \sum_{d\in G_d} R_d \geq 0 \tag{51}$$

where R_a and R_d represent the reputation values of the agreeing and disagreeing microgrids, respectively. This mechanism ensures that high-reputation microgrids have greater influence, preventing low-credibility nodes from dominating decisions through simple majority.

Traditional consensus mechanisms, such as majority voting and participation-based schemes, often overlook the historical behavior and credibility of market participants. This limitation allows low-reputation microgrids to disproportionately influence trading outcomes, potentially leading to market inefficiencies and manipulation. In contrast, the proposed reputation-based consensus mechanism incorporates both historical performance and reputation scores, assigning greater decision-making weight to trustworthy microgrids. This approach mitigates the influence of unreliable participants, enhances transaction fairness, and supports long-term market stability. Key advantages of the reputation-based consensus mechanism include:

- **Resilience to Dishonest Behavior:** Limits the influence of microgrids with a history of unreliable or non-compliant transactions.
- **Resistance to Market Manipulation:** Prevents unreliable participants from exploiting the system through frequent but low-quality engagement.
- **Promotes Long-Term Stability:** Encourages sustained and responsible participation by granting more decision-making power to reputable microgrids, discouraging short-term opportunism.

By incorporating a reputation-based consensus mechanism, this study presents an innovative solution for cross-microgrid energy trading that enhances fairness, transparency, and efficiency. Unlike conventional consensus approaches that assign equal voting weight to all participants regardless of their historical performance, the proposed mechanism introduces reputation-weighted decision-making. Transactions are not determined solely by majority agreement, but by the aggregated reputations of the agreeing and disagreeing microgrids. As a result, microgrids with demonstrated reliability and positive transaction histories exert greater influence in the consensus process. This prioritization fosters a more trustworthy and equitable trading environment, ultimately enhancing market integrity, operational efficiency, and overall system stability.

(4) Game-Theoretic Interpretation

From a game-theoretic perspective, the dynamic matching and reputation-based consensus mechanisms jointly form a repeated game framework where microgrids iteratively interact to maximize their long-term utility.

- Dynamic Matching as a Cooperative Game: The stable matching process based on preference lists can be interpreted as a form of cooperative game, where each microgrid aims to maximize its utility by selecting trading partners with favorable prices and reputation levels. The deferred acceptance algorithm(The modified Gale-Shapley algorithm in Section 5.2.3 - (2)) we use ensures stability (i.e., no blocking pairs), which is aligned with the core solution concept in cooperative games.
- Reputation-Based Consensus as a Repeated Game: The consensus validation process based on reputation can be viewed as a repeated game where microgrids are incentivized to behave honestly over multiple trading rounds. High-reputation nodes gain more influence in the consensus mechanism, while malicious behavior is penalized through reduced future influence. This aligns with the principles of evolutionary game theory and fosters cooperative behavior over time.

Each participant's strategy—ranging from truthful bidding to cooperative behavior—is influenced by its expected future gains, which are in turn affected by its reputation score. The use of reputation as a cumulative payoff mechanism induces incentive compatibility: microgrids are motivated to behave honestly and efficiently to gain higher reputational influence in future trading rounds.

Furthermore, the matching process resembles a dynamic cooperative game, where stable matchings emerge as Nash equilibria under the influence of utility-driven preference rankings.

Future work may formalize these interactions using evolutionary game theory or mechanism design to analyze equilibrium strategies and convergence under incomplete information.

5.3. Price adjustment strategy

Building on the basic tradable conditions and the consensus mechanism, trading requests that remained unmatched in the previous round undergo price adjustments before entering the next round of matching.

Traditional price adjustment mechanisms often rely on fixed-value strategies, which, while simple, lack adaptability in dynamic multi-party trading environments. Such rigidity can lead to inefficient pricing, especially under conditions of supply-demand imbalances and heterogeneous participant behavior, thereby impeding transaction efficiency.

To address these limitations, this paper proposes a hybrid price adjustment strategy that combines fixed-value and dynamic adjustments. Initially, fixed-value adjustments ensure market stability and avoid excessive fluctuations. Subsequently, the mechanism shifts to dynamic adjustment, where prices adapt in response to current supply-demand conditions and MEI. This adaptive approach captures more complex market dynamics, improves pricing accuracy, enhances matching efficiency, and promotes fairness by better reflecting actual market conditions.

5.3.1. The "3-k adjustment strategy"

To further enhance pricing adaptability, transaction efficiency, and market stability, we propose a dynamic price adjustment mechanism that integrates the MEI and the market SDR. This mechanism follows the "$3-k$ Adjustment Strategy", which combines fixed-value adjustments in the initial rounds with dynamic adjustments in later stages to optimize pricing and improve transaction success rates.

(1) Fixed-value adjustment

The first three rounds adopt a fixed-value adjustment approach to maintain stability and assess market responses, while subsequent k rounds utilize dynamic adjustments based on MEI and supply-demand conditions to refine pricing strategies.

- First round: A minor price adjustment is applied to maintain market stability while observing initial market reactions.
- Second round: If a match is not achieved in the first round, the adjustment magnitude is increased to accelerate convergence toward a potential matching point.
- Third round: If no transaction has occurred, a more aggressive price adjustment is introduced to facilitate as many matches as possible.

The primary objective of the price adjustment mechanism is to increase buyers' bids while reducing sellers' asking prices, thereby narrowing the bid-ask spread and improving the likelihood of successful transactions. During the first three rounds, price adjustments follow a tiered approach, formulated as follows:

$$p_s^k = p_s^{k-1}(1-\mu) \tag{52a}$$

$$p_b^k = p_b^{k-1}(1+\mu) \tag{52b}$$

Where p_s is the seller's price, p_b is the buyer's price, and μ is a price adjustment factor.

By incorporating both fixed and dynamic price adjustments, the "$3-k$ Adjustment Strategy" ensures that price modifications are adaptive, efficient, and responsive to market conditions, ultimately improving matching success rates and enhancing overall market fairness and stability.

(2) Consideration of supply and demand

The SDR is defined as the ratio of total market supply to demand:

$$SDR = \frac{S_e}{D_e} \tag{53}$$

where S_e and D_e represent the supply and demand of the multi-microgrid market, respectively.

To ensure market efficiency, price adjustments are made based on the supply-demand balance. When supply exceeds demand($SDR > 1$), the seller's price adjustment factor is modified as follows:

$$f(SDR)_s = \frac{SDR-1}{SDR} \tag{54}$$

When supply is less than demand($SDR < 1$), the buyer's price adjustment factor is adjusted accordingly:

$$f(SDR)_b = \frac{1-SDR}{SDR} \tag{55}$$

The general supply-demand adjustment function is formulated as:

$$f(SDR)_s = \begin{cases} \dfrac{SDR-1}{SDR}, SDR > 1 \\ 0, SDR \le 1 \end{cases} \tag{56a}$$

$$f(SDR)_b = \begin{cases} 0, SDR \ge 1 \\ \dfrac{1-SDR}{SDR}, SDR < 1 \end{cases} \tag{56b}$$

The corresponding price adjustment functions are defined as:

$$p_s^k = p_s^{k-1}(1 - f(SDR)_s) \tag{57a}$$

$$p_b^k = p_b^{k-1}(1 + f(SDR)_b) \tag{57b}$$

where p_s^k and p_b^k represent the seller's and buyer's adjusted prices in the k-th round, respectively. $f(SDR)_s$ and $f(SDR)_b$ are the supply-demand-dependent price adjustment factors.

(3) Consideration of multi-dimensional evaluation index

Incorporating the MEI ensures that price adjustments reflect the credibility and reliability of microgrid participants. Microgrids with high MEI values undergo smaller price adjustments, whereas those with lower MEI values experience larger adjustments to better align with market conditions. The MEI-dependent price adjustment factor is formulated as:

$$f(MEI) = 1 - MEI \tag{58}$$

The resulting price adjustment functions are:

$$p_s^k = p_s^{k-1}(1 - f(MEI)_s) \tag{59a}$$

$$p_b^k = p_b^{k-1}(1 + f(MEI)_b) \tag{59b}$$

where $f(MEI)_s$ and $f(MEI)_b$ are the price adjustment factors determined by the MEI of the seller and buyer, respectively.

(4) Comprehensive adjustment factor

To achieve a balanced and adaptive pricing strategy, we introduce a comprehensive price adjustment factor (μ^k) that integrates both the SDR-based and MEI-based adjustment functions:

$$\mu^k = \beta_1\ f(SDR) + \beta_2\ f(MEI) \tag{60}$$

where β_1 and β_2 are weighting coefficients that determine the relative influence of supply-demand conditions and multi-dimensional evaluation on price adjustments.

Using this comprehensive adjustment factor, the seller's and buyer's prices are updated as follows:

$$p_s^k = p_s^{k-1}(1 - \mu^k) \tag{61a}$$

$$p_b^k = p_b^{k-1}(1 + \mu^k) \tag{61b}$$

To prevent extreme price fluctuations, we introduce upper and lower bounds on the adjustment factor:

$$\mu = clip(\mu^k, \mu_{min}, \mu_{max}) \tag{62}$$

where μ_{min} is the minimum adjustment range, ensuring a minimum adjustment level, and preventing stagnation even for high-MEI microgrids. μ_{max} is the maximum adjustment range, which limits excessive price fluctuations for low-MEI participants, preventing instability in the market.

The final bounded price adjustment function is:

$$p_s^k = p_s^{k-1}(1 - \mu) \tag{63a}$$

$$p_b^k = p_b^{k-1}(1 + \mu) \tag{63b}$$

5.4. Multi-microgrid trading process

When M microgrids participate in a multi-microgrid transaction, the trading process follows a structured sequence of steps to ensure efficient, fair, and reliable transactions. The detailed process is as follows:

(1) Bid Information Submission: Each microgrid submits its power supply/demand and initial price, represented as $B_{MG_i} = \{q_{net}, p_{init}\}$, where q_{net} is the net power quantity and p_{init} is the initial bid price. This information is broadcast to all other participating microgrids.

(2) Preference List Generation: Each microgrid receives a set of bid information, denoted as $B_t = \{B_{MG_1}, \dots, B_{MG_k}\}$, along with the geographical distance data of other participating microgrids. Based on this information, each microgrid computes the distance-based price for all potential trading counterparts. These values are then used to rank feasible trading partners, taking both tradability conditions and geographic proximity into account. The result is an available matching preference list, denoted as $\Theta_{\text{avail}} = \{MG_1, \dots, MG_m\}$, which reflects the prioritized order of potential transaction partners. Once generated, each microgrid broadcasts its preference list to the network to facilitate distributed matching.

(3) Transaction Matching Calculation: Each microgrid receives the preference lists from other microgrids and applies the dynamically adjusted bidirectional stable matching algorithm to compute optimal transaction matches, represented as $\mathcal{M}^{(op)} = \{tx_1, \dots, tx_k\}$.

(4) Update and Price Adjustment: After completing a matching round, the following updates occur:

1. Update the MEI for the successfully matched microgrids;
2. Adjust transaction prices based on the proposed dynamic adjustment mechanism;
3. Update supply and demand information for the next matching round;
4. Repeat from Step 2 for subsequent rounds of matching.

(5) Trading with the Utility Grid: If all consumers (or all producers) still have unmatched transactions after multiple rounds, they will trade directly with the utility grid($\mathcal{M}^{(UG)}$) to ensure energy balance and market continuity.

(6) Definitive Matching Result Generation: Once multiple rounds of matching are completed, the set of successful transactions is determined, represented as $\mathcal{M}^{(def)} = \mathcal{M}^{(op)} + \mathcal{M}^{(UG)}$. This definitive matching result is then broadcasted to all microgrids.

(7) Verification of Matching Results: Each microgrid receives the definitive matching result from other microgrids and verifies it using

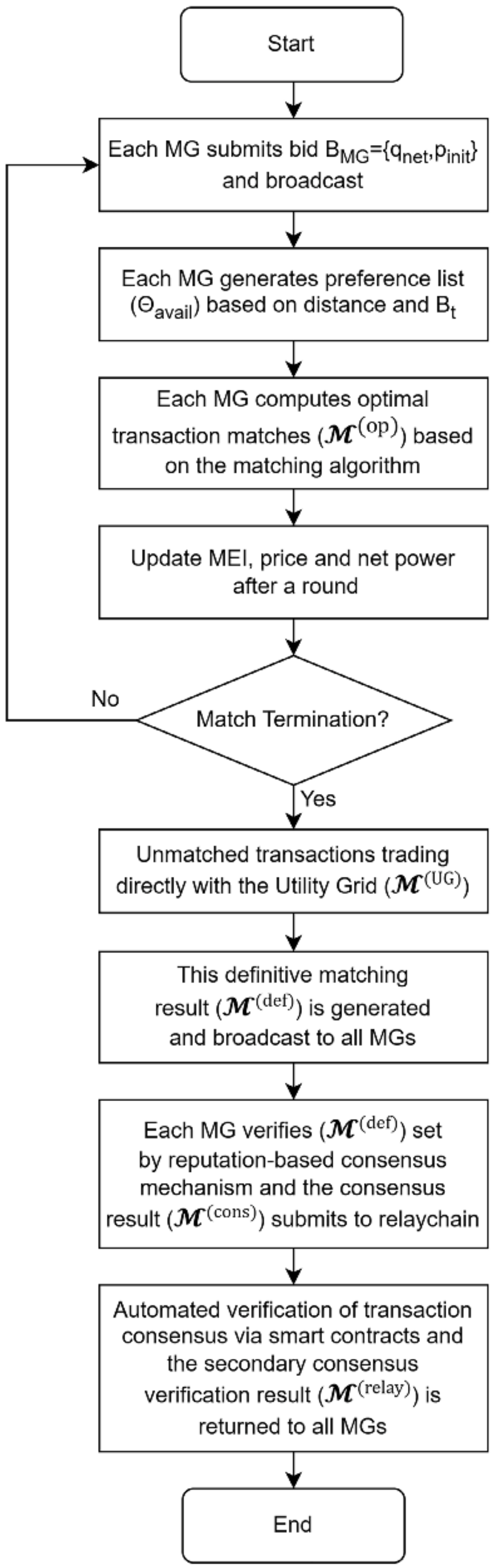

Fig. 3. Flowchart of multi-microgrid trading process.

the reputation-based consensus mechanism. The verified consensus result $\mathcal{M}^{\text{cons}}$ is then generated. All participating microgrids submit this consensus verification result to the relay chain for further validation.

(8) Secondary Consensus Verification and Result Storage: Automated verification of transaction consensus via smart contracts.

1. Finalizing the consensus verification result $\mathcal{M}^{\text{relay}}$, which is then returned to all microgrids.
2. Recording the final verification result in the relay chain to ensure transparency and immutability.
3. Updating the reputation scores of the participating microgrids based on their transaction history and consensus compliance.

The overall multi-microgrid trading process is shown in the Fig. 3.

This structured multi-microgrid trading process ensures efficient transaction execution, dynamic price adaptability, and fair trading opportunities. By integrating preference-based matching, dynamic price adjustments, and a reputation-based consensus mechanism, the proposed approach enhances market stability, transaction reliability, and trustworthiness within the decentralized energy trading ecosystem.

The convergence of the proposed price adjustment and matching mechanism is theoretically proven in Appendix A.

5.5. Handling uncertainty for cross-microgrids trading

While traditional uncertainty modeling techniques such as SP and RO are commonly applied within single microgrid contexts, cross-microgrid trading introduces additional layers of uncertainty propagation due to interconnected energy flows and decision feedback loops. Our proposed framework does not explicitly rely on probabilistic uncertainty models at the inter-microgrid level, but instead incorporates a series of mechanism designs that implicitly enhance robustness and adaptability to uncertain conditions.

Specifically, our framework integrates three synergistic components to manage uncertainty in cross-microgrid trading:

Dynamic Price Adjustment Strategy: Prices are adaptively updated based on market supply-demand. When the system detects market disequilibrium—price adjustment factors are adjusted to incentivize corrective actions. This dynamic adjustment functions as a self-regulating mechanism, mitigating the impact of short-term fluctuations and improving overall market responsiveness.

Reputation-Driven Transaction Consensus: To address potential non-performance or delivery failures due to uncertainty, a delayed verification and reputation penalty mechanism is introduced. Microgrids that fail to honor commitments face reputation deductions, which in turn affect their future participation and matching priority. This strategy discourages opportunistic behavior, promotes long-term reliability, and ensures that transaction decisions are guided by trustworthy participants.

Multi-Dimensional Evaluation Model: The MEM aggregates multiple indicators—including environmental contribution, transaction success, and bidding rationality—to quantify a microgrid's stability and reliability. Microgrids with stronger performance under uncertainty naturally achieve higher evaluation scores, leading to more favorable matching outcomes. This selection bias effectively steers the market toward robust participants, increasing trading efficiency and resilience. Additionally, it facilitates trade matching between microgrids by influencing price adjustments.

In summary, although uncertainty is not explicitly modeled at the inter-microgrid level, our framework embeds robustness through pricing, reputation, and evaluation dynamics, offering a practical and scalable approach to resilient P2P trading. These mechanisms enable the system to indirectly absorb, mitigate, and adapt to uncertainty without relying on centralized forecasting or predefined probability distributions, fostering a self-correcting, reputation-sensitive, and information-adaptive market that ensures reliable cross-microgrid energy trading under volatile renewable generation and uncertain demand conditions.

While our proposed mechanisms address uncertainty implicitly through design features such as dynamic pricing and reputation feedback, it is also important to position our work relative to existing models that explicitly incorporate uncertainty through stochastic or robust optimization. A comparative discussion is provided below.

5.6. Comparison with uncertainty-aware models: limitations and integration potential

In order to contextualize the limitations and potential extensions of our deterministic design, we provide a comparative discussion with representative uncertainty-aware models.

Existing approaches address uncertainty in P2P energy markets through different strategies: Guo et al. [13] used chance-constrained programming for probabilistic reserve allocation, requiring accurate uncertainty modeling; Zhang et al. [32] adopted data-driven DRO to handle worst-case deviations at the cost of computational complexity and potential conservatism; while Mansouri et al. [33] integrated robust optimization into an ADMM framework, ensuring security against worst-case scenarios but potentially compromising economic efficiency. These methods demonstrate trade-offs between robustness, efficiency, and computational tractability in uncertainty-aware market designs.

Compared to existing uncertainty-aware models, our study focuses on developing a cross-chain trading system for microgrids that prioritizes three key features: smooth interoperability between different systems, truly decentralized operation, and reliable trust-building among participants. We've chosen to use deterministic assumptions in this initial phase to clearly assess how well our blockchain-based coordination method works on its own, without the added complexity of uncertainty factors. This focused approach helps us understand exactly how our design innovations affect system performance. While this gives us clear insights into our core mechanism, we recognize it doesn't account for real-world uncertainties, which is an important limitation we plan to address in future work. The main contribution of this paper is creating a solid foundation for cross-chain microgrid trading that can later be combined with uncertainty-handling methods.

6. Cross-chain system implementation for multi-microgrids energy trading

This section presents the implementation of a cross-chain system for multi-microgrid energy trading. The proposed architecture ensures secure, efficient, and transparent energy transactions between microgrids through blockchain and cross-chain technologies.

6.1. Cross-chain system architecture

The system is structured as a two-layer microgrid power trading architecture, as illustrated in Fig. 4, comprising:

1. The microgrid blockchain layer (lower layer): Handles local energy trading within individual microgrids.
2. The cross-microgrid relay chain layer (upper layer): Facilitates energy trading across multiple microgrids.

At the microgrid blockchain layer, DERs are integrated into a blockchain platform to enable secure and transparent local energy trading. Smart contracts automate transaction price negotiations, while all transaction clearing results are immutably recorded on the blockchain.

Each microgrid operates its own blockchain, ensuring autonomy and security. These individual blockchains are interconnected via cross-chain technology, enabling energy transactions across multiple microgrids. The relay chain layer acts as a decentralized cross-chain communication infrastructure, enabling secure interoperability between independently operating microgrid blockchains. Each microgrid blockchain is registered on the relay chain by submitting its chain

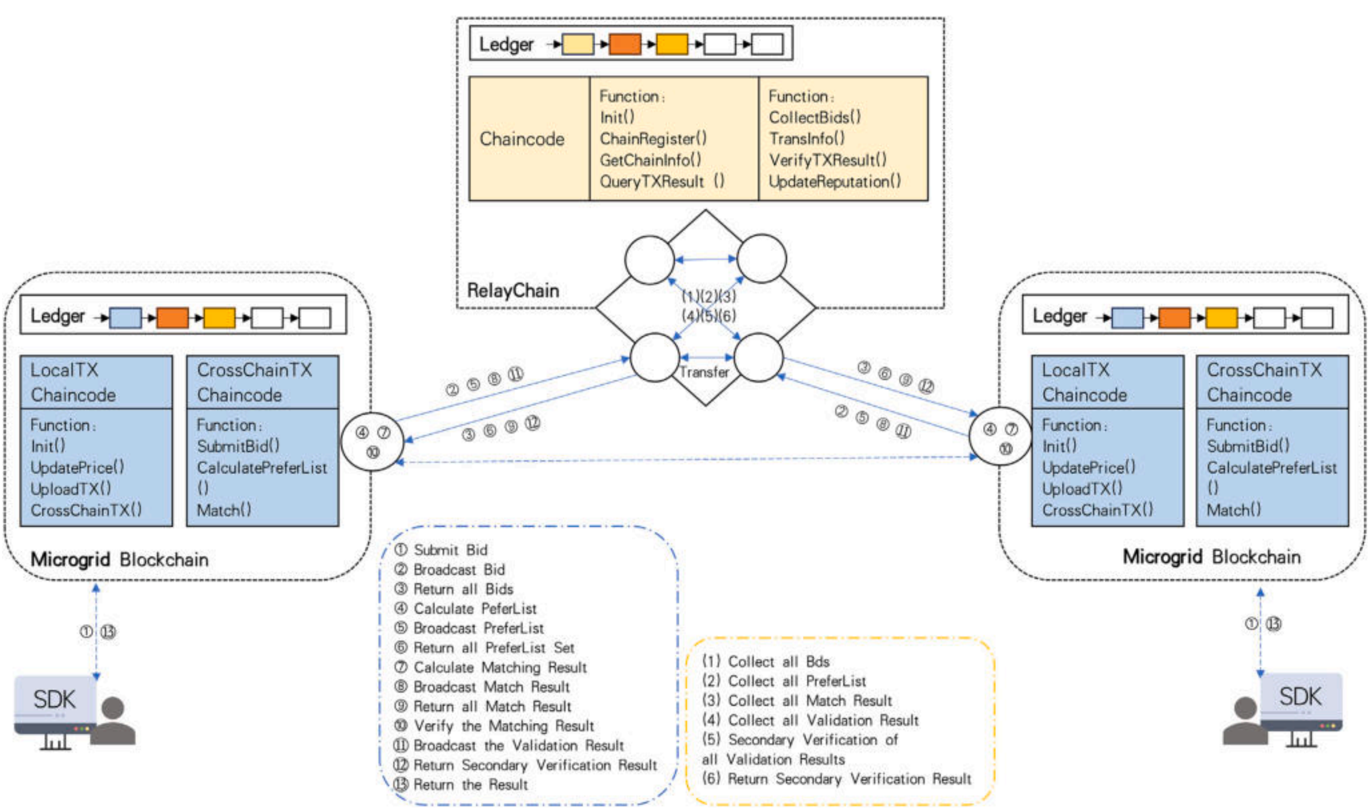

Fig. 4. The cross-chain system architecture.

identity (ChainID), public-private key credentials, and microgrid information. Once registered, the relay chain enables a structured process of cross-chain interaction, including:

(1) the collection and aggregation of energy bids (B_t),
(2) the exchange of preference lists (Θ_{avail}),
(3) the coordination of dynamic decentralized matching ($\mathcal{M}^{\text{def}}$),
(4) and the final secondary consensus verification process ($\mathcal{M}^{\text{relay}}$).

All operations are executed by smart contracts deployed on the relay chain, ensuring auditability, immutability, and removing the need for centralized validators. By embedding a reputation-based weighted consensus protocol within the relay chain, the system guarantees that high-reputation microgrids exert greater influence during inter-chain transaction validation, thereby enhancing trust and system resilience. Furthermore, this cross-chain architecture is designed to be blockchain-agnostic and can be extended to support various blockchain frameworks, making it highly adaptable to heterogeneous deployment environments.

This architecture enables seamless, secure, and verifiable cross-microgrid energy trading by integrating synchronized bidding, matching, and validation processes into a decentralized relay chain infrastructure.

6.2. Internal trading implementation

Within a single microgrid, participants engage in local energy trading through a blockchain-based process. The specific steps are as follows:

(1) Transaction Initialization: Participants confirm their energy supply/demand and provide relevant data ($data = \{\alpha, \beta, \gamma, \underline{E}, \overline{E}, \dots\}$). They connect to the blockchain platform via the SDK and submit transaction requests.
(2) Negotiation and Price Agreement: Participants use distributed optimization algorithms and smart contracts to negotiate transaction terms. The agreed price is securely updated and recorded on the blockchain.
(3) Transaction Confirmation: Once the transaction is finalized, the clearing results are permanently recorded on the blockchain, ensuring data integrity and immutability.

6.3. External trading implementation

When internal transactions within a microgrid remain unmatched, cross-microgrid trading is initiated using the relay chain. The process consists of two key components, Microgrid Blockchain Process and Relay Chain Process.

6.3.1. Microgrid blockchain process

When there is an imbalance in transactions within the microgrid, the microgrid blockchain will connect to the relay chain to conduct external transactions across the microgrid. When external transactions occur, the specific process of the microgrid blockchain is as follows:

(1) Cross-chain Initialization: During the cross-chain initialization phase, each microgrid blockchain registers with the relay chain by submitting the following information: node identity ($NodeID$), public key ($PubKey$), organizational certificate ($Cert$), and basic participant information ($Info = \{IP, Org, Email, Addr, \dots\}$). Public-private key verification is performed before each cross-chain operation to ensure secure data transmission and storage.
(2) Bid Submission and Broadcasting: Each microgrid aggregates its unfulfilled energy bids, denoted as $B_{MG_i} = \{q_{net}, p_{init}\}$—representing the net supply/demand and corresponding initial price—and broadcasts this information to other microgrid blockchain networks via the relay chain, thereby facilitating subsequent cross-microgrid energy trading.
(3) Preference List Generation and Broadcasting: Upon receiving the bid set($B_t = \{B_{MG_1}, \dots, B_{MG_k}\}$) from other microgrid blockchains, each microgrid generates its own preference list($\Theta_{avail} = \{MG_1, \dots, MG_m\}$) based on the bid information and geographical distance data. The generated Θ_{avail} is then broadcast via the relay chain and exchanged among participating microgrid blockchains to facilitate cross-chain matching.
(4) Transaction Matching: Each microgrid receives preference lists from other microgrid blockchains. The dynamic bidirectional stable matching algorithm is applied to compute definitive transaction matching result($\mathcal{M}^{def}$). The definitive matching results are then broadcast to other microgrid blockchains via the relay chain.
(5) Transaction Verification: Each microgrid verifies the definitive matching results using the reputation-based consensus mechanism. The verified consensus results($\mathcal{M}^{cons}$) are submitted to the relay chain for secondary consensus validation.
(6) Consensus Verification and Storage: The relay chain performs secondary consensus verification and returns the verified result($\mathcal{M}^{relay}$) to the microgrid blockchain. The verified result is stored on the microgrid blockchain, ensuring permanent record-keeping.

6.3.2. Relay chain process

When conducting cross-microgrid external transactions, as the communication bridge for cross-microgrid transactions, the relay chain performs the following operations:

(1) Microgrid Registration: Receives microgrid relevant information and security certificates from the microgrid blockchain. Uses cryptographic encryption to securely store participant information and assigns a unique ID($ChainID$) to each microgrid. Implements an access control mechanism to ensure that only registered microgrids can participate in transactions, enhancing privacy and security.
(2) Bid Collection and Broadcasting: Aggregates bids from all microgrid blockchains. Broadcasts the aggregated bids($B_t = \{B_{MG_1}, \dots, B_{MG_k}\}$) to all microgrid blockchains.
(3) Preference List Collection and Broadcasting: Collects preference lists(Θ_{avail}) from each microgrid blockchain. Broadcasts the preference lists to all participating microgrid blockchains.
(4) Definitive Matching Result Collection and Broadcasting: Collects definitive matching results($\mathcal{M}^{def}$) from each microgrid blockchain. Broadcasts the $\mathcal{M}^{def}$ to all microgrid blockchains.
(5) Secondary Consensus Verification and Reputation Updates: Receives consensus verification results($\mathcal{M}^{cons}$) from each participating microgrid. Performs secondary consensus verification on these results using reputation-based consensus mechanism. Returns the secondary consensus verification result($\mathcal{M}^{relay}$) to all microgrids and records the result on the relay chain. Updates the reputation scores of microgrids based on their transaction behavior and consensus compliance.

Through cross-chain technology, the blockchains of individual microgrids communicate with the relay chain, aggregating and transmitting transaction bid information. This enables data sharing and synchronization across microgrids. Using smart contract technology, transactions are automatically executed, and power transaction results and settlement information are recorded on the blockchain, ensuring that cross-microgrid transactions remain transparent and secure.

6.4. Comparison with classical consensus mechanisms

To evaluate the effectiveness of the proposed reputation-weighted consensus mechanism, we compare it with classical consensus protocols commonly adopted in blockchain-based systems, including Proof

Table 2
Comparison of consensus mechanisms.

Criterion	PoW	PoS	BFT	Reputation-Based (Ours)
Energy Efficiency	Low	High	High	High
Manipulation Resistance	Strong	Moderate (stake attacks)	Strong	Strong (trust history)
Fairness	Low (hash rate bias)	Low (wealth bias)	Medium	High (behavior-based weighting)
Scalability	Low	Medium	Low	High
Suitability for Energy Trading	Poor	Limited	Limited (small networks)	Excellent
Computation Overhead	High	Low	Low	Low
Trust Foundation	Hash power	Token stake	Node identity	Historical reputation

of Work (PoW), Proof of Stake (PoS), and Byzantine Fault Tolerance (BFT). Each protocol exhibits unique characteristics in terms of security, efficiency, and applicability. However, in the context of decentralized multi-microgrid energy trading, the system design requires not only robustness and resistance to manipulation, but also low computational overhead and scalability to heterogeneous microgrid blockchain environments.

Table 2 provides a structured comparison across several criteria: energy efficiency, manipulation resistance, fairness, scalability, and suitability for energy trading. As shown, PoW offers strong security but incurs high energy costs, making it unsuitable for energy-constrained microgrid systems. PoS reduces computational demand but may suffer from wealth concentration and stake manipulation. BFT is efficient and secure in small-scale environments but does not scale well to large or open networks. In contrast, our reputation-weighted consensus mechanism achieves a balance by assigning dynamic influence to participants based on historical trading behavior, thus enhancing trust, resisting manipulation, and ensuring high compatibility across microgrid blockchain environments.

In summary, the reputation-weighted consensus mechanism aligns well with the needs of cross-chain microgrid energy trading: it minimizes computation and energy costs, discourages malicious behavior, promotes long-term fair participation, and scales to large heterogeneous networks. These characteristics make it a promising alternative to conventional consensus mechanisms in energy applications.

7. Simulation results

7.1. Basic setup

To evaluate the effectiveness of the proposed multi-microgrid energy trading mechanism, a simulation experiment was conducted. The cross-chain system implementation for multi-microgrid energy trading consists of six independent blockchains, each representing a separate microgrid, and one relay blockchain that facilitates inter-chain communication. The blockchain system is implemented using Hyperledger Fabric 2.2.9, deployed on virtual machines running Ubuntu 18.04. Smart contracts (chaincodes) for energy trading were installed on the blockchain and executed using the Go SDK, which enables seamless interaction with the blockchain system.

The multi-microgrid system consists of five interconnected microgrids, each containing 10 agents deployed on the IEEE 9-bus test system. The test system integrates conventional generators, consumer loads, wind power generators and BESS, simulating a realistic distributed energy environment. The experimental platform is configured in Windows 11 with a 12th Gen Intel® Core™ i7-12700 CPU (2.10 GHz) processor, and 32 GB RAM. For local microgrid market optimization, the convex optimization problem is solved using the Gurobi 12.0 solver, ensuring efficient and precise transaction calculations.

Remark:Each microgrid internally adopts the IEEE 9-bus system topology for simulating local power flow and resource allocation. While the IEEE 9-bus system is traditionally used for transmission-level studies, it provides a well-established benchmark with transparent data and validated models. In this context, we use it solely as a simulation framework to test the power flow-aware energy trading algorithms within each microgrid. The voltage levels are conceptually adjusted to align with distribution-level operation, and the focus remains on validating the intra- and inter-microgrid trading logic rather than replicating exact physical conditions.

Table 3
Agent's parameters in single microgrid.

Agent	Bus	α	β	γ	$\underline{E}$	$\overline{E}$
C1	1	0.027	15.181	5.1	6.0	60.0
C2	2	0.033	18.483	5.1	5.5	55.0
C3	3	0.043	12.835	5.1	5.0	50.0
U1	1	0.022	12.737	5.1	−40.0	−4.0
U2	5	0.023	13.889	5.1	−45.0	−4.5
U3	7	0.034	15.712	5.1	−45.0	−4.5
U4	9	0.035	12.044	5.1	−35.0	−3.5
W1	5	0.027	5.286	5.1	0.0	30.0
W2	7	0.026	6.220	5.1	0.0	25.0
B1	9	0.017	10.810	5.1	0.0	50.0

7.2. Convergence performance

7.2.1. Convergence performance of single microgrid

Firstly, we test a single microgrid market convergence performance, which has 10 agents on the IEEE 9-bus test system consisting of 3 conventional generators, 4 users, 2 wind power generators, and 1 BESS. Agents C1-C3 are conventional generators, agents U1-U4 are users, W1-W2 are wind power generators, and B1 is BESS. The parameters of agents are summarized in Table 3. In addition, the bilateral trade coefficient c_{nm} varies from 0.3 to 0.7, $\eta_c = \eta_{dc} = 0.95$, $\underline{SoC} = 0.2$, $\overline{SoC} = 0.8$, $Q_B = 100$ kWh, $\Delta_t = 1\ h$, the initial value $SoC = 0.5$, and the resistance and reactance between nodes are set to 0.2 and 1, all global stopping criteria are chosen to be 10^{-3}, and the penalty factors ρ and τ are set to 1.

The final trading results are presented in Table 4. It can be observed that P2P energy trading within a single microgrid achieves convergence, ensuring that all energy demand is satisfied. The observed price variations among agents reflect the product differentiation effect, where different participants trade at slightly different prices based on their preferences, locations, and energy availability. Fig. 5 illustrates the convergence performance that is measured by the global residuals defined in (26)-(27) in a single microgrid scenario, which represents the evolution of the primal residual and dual residual over iterations. The y-axis is plotted on a logarithmic scale to better visualize the residual reduction trend. The horizontal dashed line indicates the convergence threshold. The primal and dual residuals decrease progressively as the iteration count increases. Initially, the residual values are high, but they experience an exponential decline, reaching a stable state after 200 iterations. This indicates that the optimization algorithm effectively converges, reducing residual errors to acceptable levels. Compared to conventional centralized energy trading, the P2P approach demonstrates faster convergence lower residual errors ensures price flexibility and enhances local market efficiency, reducing reliance on the central utility grid.

Table 4
The trading results of P2P energy trading in a single microgrid.

Agent	U1	U2	U3	U4
C1	0.58 kW/12.18$/kW	2.96 kW/12.21$/kW	1.94 kW/12.22$/kW	0.52 kW/12.17$/kW
C2	/	5.50 kW/12.26$/kW	/	/
C3	/	3.47 kW/12.27$/kW	/	1.53 kW/12.25$/kW
W1	1.73 kW/11.97$/kW	12.87 kW/12.05$/kW	13.96 kW/12.05$/kW	1.45 kW/12.02$/kW
W2	/	10.04 kW/12.03$/kW	14.97 kW/12.03$/kW	/
B1	4.24 kW/12.06$/kW	10.16 kW/12.12$/kW	14.11 kW/12.09$/kW	/

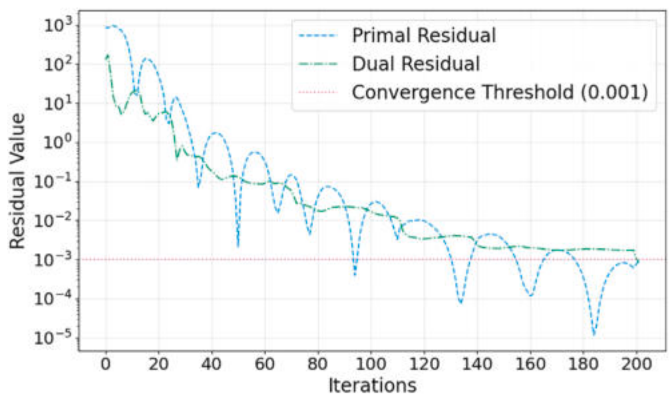

Fig. 5. Primal and dual residuals convergence process in a single microgrid.

7.2.2. Convergence performance of multi-microgrid

The proposed method is applied to test the convergence of multi-microgrid energy trading. The multi-microgrid system consists of five microgrids, whose key parameters are listed in Table 5. Fig. 6 illustrates the process in which the microgrid cannot converge. When the residual change value is less than the given threshold, the iteration will stop. It can be seen that the iteration stops automatically and does not continue indefinitely. Some microgrids exhibit energy surplus, while others experience energy deficits, enabling cross-microgrid transactions. The buying price from the utility grid is set at $PBU = 14.3$, while the selling price to the utility grid is set at $PSU = 7.2$, ensuring an incentive for direct transactions. The price adjustment factor is constrained between $\mu_{min} = 0.1$ and $\mu_{max} = 0.3$, preventing excessive fluctuations while allowing adaptive price tuning. The physical distance matrix (Equation (64)) accounts for geographical distances, ensuring realistic distance differences.

The final multi-microgrid trading outcomes are presented in Table 6. The results indicate that the proposed method enables convergence within 6 iterations, significantly improving efficiency compared to traditional bilateral trading, which often requires more iterations.

$$D_{phy} = \left[\begin{array}{c|ccccc} MG & 1 & 2 & 3 & 4 & 5 \\ \hline 1 & 0 & 15 & 6 & 7 & 15 \\ 2 & 15 & 0 & 14.2 & 10 & 8 \\ 3 & 6 & 14.2 & 0 & 10 & 10 \\ 4 & 7 & 10 & 10 & 0 & 14.2 \\ 5 & 15 & 8 & 10 & 14.2 & 0 \end{array}\right] \tag{64}$$

7.3. Comparison of social welfare under different matching mechanisms

This section presents the evaluation of social welfare under different energy trading mechanisms in multi-microgrid markets. Different matching mechanisms will affect social welfare in multi-microgrid energy trading. Specifically, we evaluate and compare the proposed method against the following mechanisms:

Table 5
Microgrid's parameters in multi-microgrid energy trading.

Parameters	q_{net}	p_{init}	Composed	R
MG1	−2.5	11.67	3 C,3U,4 W	15.30
MG2	−1.5	10.7	3 C,4U,3 W	8.60
MG3	0.68	8.81	3 C,3U,4 W	21.50
MG4	1.27	10.27	3 C,4U,2 W,1B	7.40
MG5	0.35	12.55	3 C,3U,4 W	5.50

Table 6
The transaction results of multi-microgrid energy trading.

Buyer	Seller	Quantities	Price	Round	WB	WS
MG1	MG3	0.68	10.21	1	2.78	2.04
MG1	MG4	1.27	11.29	4	3.82	5.20
MG2	MG5	0.35	12.19	5	0.74	1.75
MG1	UG	0.55	13.59	6	0.39	/
MG2	UG	1.00	13.59	6	0.71	/
MG2	UG	0.15	14.01	6	0.04	/

- **Single Round Matching(SRM)**: Each buyer and seller can only be matched once in a single round of trading. Once a match is made, it cannot be modified or adjusted.
- **Multiple Round Matching(MRM)**: If a buyer's or seller's net power demand is not fully satisfied in the first round, they continue participating in subsequent rounds until their demand or supply is fully matched.
- **Traditional Double Auction(TDA)**: Buyers and sellers submit bids and offers, and transactions occur based on matching prices. The process alternates between buyers and sellers until all possible matches are completed.
- **Continuous Double Auction(CDA)**: An extension of the traditional double auction where price adjustments (buyers increase, sellers decrease) and dynamic supply-demand adjustments are incorporated after each trading round, allowing for more adaptive market behavior.

The results are illustrated in Fig. 7, showing the social welfare for buyers (WB), sellers (WS), and the total market (WF) under various matching mechanisms, including our proposed method, SRM, MRM,TDA, and CDA. As depicted in Fig. 7, the proposed method achieves the highest total welfare ($WF = 17.48$), demonstrating an improvement compared to the continuous double auction methods ($WF = 17.33$). While SRM provides the lowest welfare due to the lack of iterative adjustments, MRM and CDA allow for more rounds of trading, thereby increasing the welfare. However, these methods require more iterations (e.g., CDA converging in 8 rounds, MRM converging in 13 rounds and our proposed method converging in 6).

The proposed method is based on supply and demand to determine the matching initiator, who can initiate matching requests to potential transaction partners on their preference list. As depicted in Table 5, it can be seen that demand exceeds supply, which is the seller's market. The $WS = 8.99$ is relatively higher than other methods. Compared to other methods, it is significantly fairer.

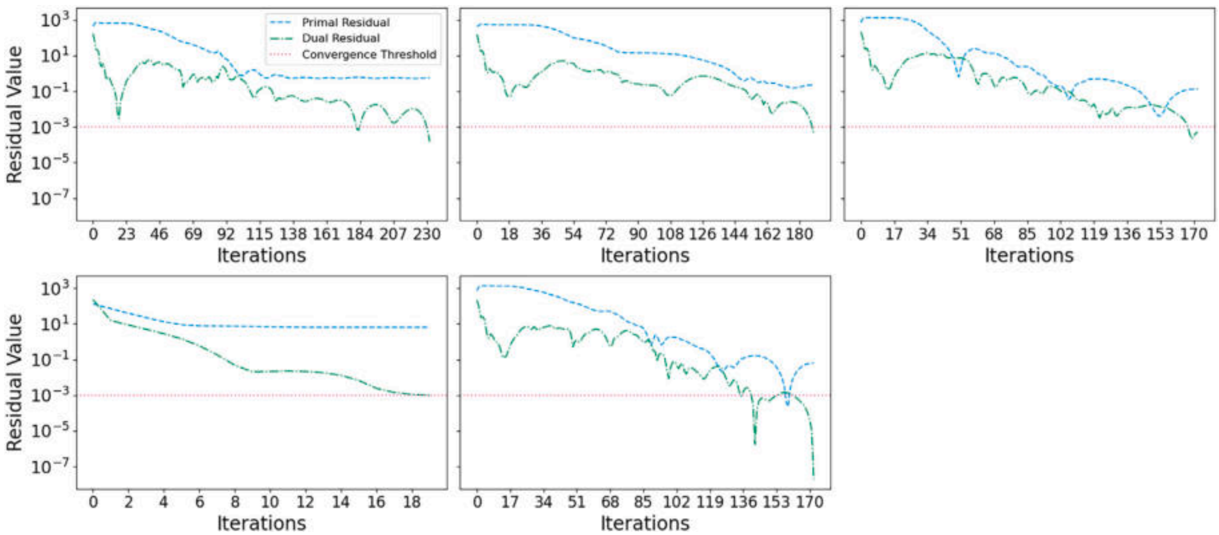

Fig. 6. Primal and dual residuals iterative process of each microgrid.

Fig. 7. Comparison of social welfare under different mechanisms.

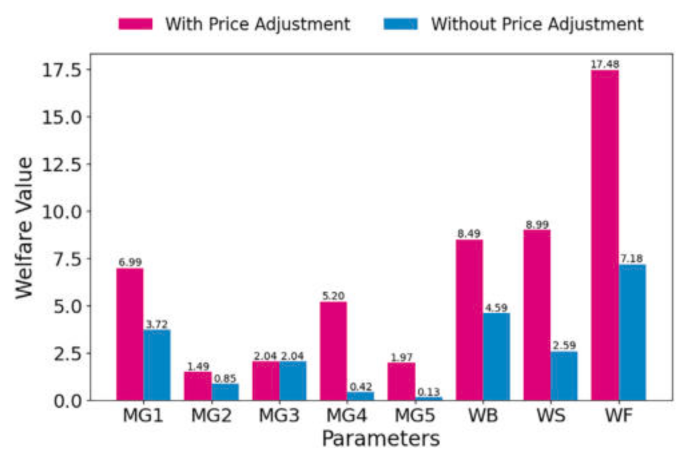

Fig. 8. Social welfare comparison of WPA and NoPA.

Through this comparative analysis, we evaluate the effectiveness of the proposed method in improving transaction efficiency and maximizing social welfare. Our method successfully balances efficiency and welfare by reaching an optimal solution in only 6 iterations. This confirms that the proposed method enhances market performance. In addition, in terms of market fairness, it can balance the social welfare of buyers/sellers well based on market supply and demand.

7.4. Effectiveness of price adjustment

To evaluate the impact of price adjustments, we compare two scenarios:

- **With Price Adjustment (WPA)**, where transaction prices dynamically update based on SDR and MEI.
- **Without Price Adjustment (NoPA)**, where prices remain fixed across trading rounds.

Fig. 8 illustrates the impact of price adjustment on social welfare. Without price adjustment, market inefficiencies arise, leading to lower transaction success rates and suboptimal resource allocation. With price adjustment, microgrids dynamically adapt their bidding strategies, achieving higher total welfare ($WF = 17.48$ compared to $WF = 7.18$ in NoPA). The results confirm that price adjustment significantly enhances transaction efficiency, market equilibrium, and overall social welfare.

7.5. Effectiveness of reputation-based consensus mechanism

To validate the effectiveness of our reputation-based consensus mechanism, we conducted simulation experiments on fraudulent behavior and delayed verification.

7.5.1. Resistance to fraudulent behavior

To validate the resistance to fraudulent behavior of our reputation-based consensus mechanism, we compare it with majority voting in the presence of inconsistent trading results.

- **Majority Voting(MV)**: Determines transaction validity based on a simple majority rule. This approach may allow low-reputation participants to manipulate decisions, potentially leading to unfair or suboptimal market outcomes.
- **Reputation-based Consensus(RBC)**: Weighs participant decisions based on their reputation scores. Higher-reputation microgrids exert greater influence on consensus, ensuring that market stability and fairness are maintained.

Table 7
Computed matching results by microgrids.

	MG1	MG2	MG3	MG4	MG5	R_a	R_d
R	15.30	8.60	21.50	7.40	5.50	–	–
TX_A	agree	disagree	agree	disagree	disagree	36.8	21.5
TX_B	disagree	agree	disagree	agree	agree	21.5	36.8

Table 8
Consensus results comparison.

Consensus mechanism	MV	RBC
Final Matching Decision	TX_B	TX_A
Agreement Reputation Sum	21.5	36.8
Against Sybil Attack	Low	High
Social Welfare (WF)	15.22	17.48

To illustrate the impact of different consensus mechanisms, we simulate a scenario where five microgrids reach conflicting transaction matching results: (1) MG1 and MG3 compute matching result TX_A using the proposed method;(2) MG2, MG4, and MG5 compute matching result TX_B using the TDA method. The computed matching results and reputation values of microgrids are shown in Table 7.

Table 8 summarizes the final decisions under different consensus mechanisms.Under MV, transaction TX_B was finalized because it received support from a simple majority (3 out of 5). However, this approach does not account for reputation, allowing low-reputation microgrids to influence critical decisions, potentially leading to unfair or insecure transactions. Under RBC, reputation-weighted decisions resulted in TX_A being selected, as the total reputation score supporting TX_A ($R_a = 36.8$) exceeded that of TX_B ($R_a = 21.5$),since $\{TX_A : R_a - R_d > 0\}$ and $\{TX_B : R_a - R_d < 0\}$ based on formula (51). This approach ensures that high-reputation participants have a greater impact, improving fairness and security.

From above, there are three Key Findings. Firstly, reputation-weighted consensus mitigates Sybil attacks, where low-reputation participants could otherwise manipulate the market so that RBC enhances decision security. Besides, RBC can lead to higher social welfare and improve the market efficiency, with social welfare increasing from 15.22 (MV) to 17.48 (RBC). Finally, RBC can prevent manipulation by low-reputation microgrids, ensuring a more stable and reliable energy trading environment. These results confirm that reputation-based consensus strengthens market integrity and stability, making it a more robust alternative to traditional majority voting.

7.5.2. Simulation of delayed verification behavior

To validate the effectiveness of our reputation-based consensus mechanism under conditions of delayed verification, we conducted a simulation involving adversarial nodes that intentionally introduce delays.

To simplify the design, we designed a simulation scenario based on the microgrid and reputation data shown in Table 5. In this scenario, Microgrid 1 (MG1) is deliberately configured to delay its transaction verification in each consensus round. Specifically, MG1 postpones its response beyond the expected confirmation window($\tau_{MG1} > \tau_{reasonable}$), aiming to slow down the consensus process and disrupt the transaction timeline. For clarity and tractability, the parameters α_R, β_R,and δ_{DV} are set to 1, 10 and 5, respectively. The simulation tracks the reputation score changes of microgrid over multiple trading rounds.

As shown in Fig. 9, the reputation score of MG1 gradually decreases due to its consistently delayed behavior, which is penalized according to the proposed reputation update rule. All other microgrids have completed consensus verification on time, so their reputation values are on the rise,except for MG1. After four consecutive rounds, MG1's reputation drops below zero.

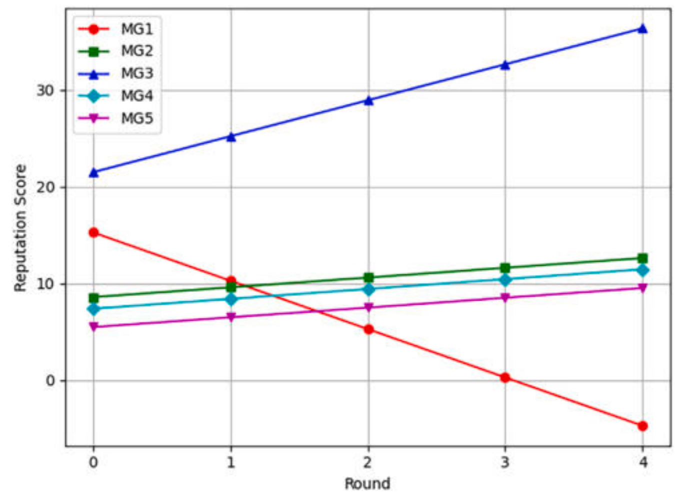

Fig. 9. Microgrid reputation evolution across rounds.

Once MG1's reputation falls below the predefined threshold, it becomes ineligible to participate in subsequent consensus rounds, effectively excluding it from the transaction process and corresponding market benefits.

These results confirm that the proposed reputation-based consensus mechanism can effectively detect and penalize delayed verification behavior, thereby preserving the security, efficiency, and fairness of the energy trading system.

7.6. Security of blockchain

In order to evaluate the impact of blockchain security on multi-microgrid energy trading, we compare two scenarios:

- **With Blockchain (WBC):** Transactions are securely recorded and verified via a cross-chain system.
- **Without Blockchain (NoBC):** Transactions occur through a centralized platform, which lacks transparency and is susceptible to manipulation.

We conduct an experiment comparing social welfare (WF) under both scenarios. The With Blockchain (WBC) transaction results are shown in Table 6. The Without Blockchain (NoBC) transaction results as calculated are shown in Table 9. The centralized trading platform has been manipulated to increase the welfare of MG2. The WBC and NoBC welfare comparison is shown in Fig. 10, which indicates that the overall welfare of market has decreased except for MG2 and harms the interests of the majority of participants.

As shown in Fig. 10 and Table 10, the WBC system significantly improves social welfare ($WF = 17.48$ vs. $WF = 11.93$ in NoBC). The blockchain-based system improves overall welfare ($WF = 17.48$) compared to the non-blockchain case ($WF = 11.93$), due to increased transparency and security. Participants are more willing to engage in energy trading, leading to better resource allocation. The results highlight that blockchain integration enhances market transparency, prevents manipulation, and increases participant trust.

7.7. Scalability of multi-microgrid energy trading

To evaluate the scalability of the proposed trading mechanism, we increase the number of microgrids participating in the market while excluding blockchain-based verification. The objective is to analyze the system's efficiency and performance in handling a growing number of microgrid participants.

We simulate a multi-microgrid energy trading market under different scales:

Table 9
The results of multi-microgrid energy trading without blockchain.

Buyer	Seller	Quantities	Price	Round	WB	WS
MG2	MG4	1.27	10.49	1	4.84	4.18
MG2	MG5	0.23	10.63	2	0.84	0.79
UG	MG5	0.12	7.56	3	/	0.04
MG1	UG	1.00	13.59	3	0.71	/
MG1	UG	1.00	14.01	3	0.29	/
MG1	UG	0.50	14.30	3	/	/
UG	MG3	0.68	7.56	3	/	0.24

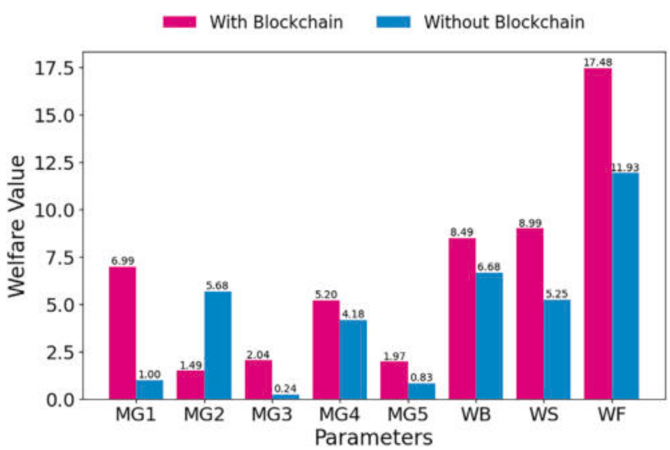

Fig. 10. Social welfare comparison of WBC and NoBC.

Table 10
Social welfare comparison: With vs. Without Blockchain.

Metrics	NoBC	WBC
Social Welfare (WF)	11.93	17.48
Transaction Security	Low	High
Market Transparency	Low	High
Matching Efficiency	High	Medium

- **Small-scale market**: 10 microgrids (5 sellers, 5 buyers)
- **Medium-scale market**: 50 microgrids (25 sellers, 25 buyers)
- **Large-scale market**: 100 microgrids (50 sellers, 50 buyers)

The trading mechanism follows the same market-clearing rules as in previous experiments, except that blockchain technology is not utilized. The distance between microgrids is defined as a random value between 1 and 10.

The results are presented in Table 11 and Fig. 11, highlighting the impact of increasing market size on runtime/computation time, transaction throughput, energy matching accuracy, and market clearing performance.

The experimental results in Table 11 show that the proposed system achieves high scalability. As the number of participating microgrids increases from 10 to 100, the runtime increases moderately while maintaining a reasonable computational cost under 100 milliseconds. Despite slight throughput degradation, the system consistently delivers high energy matching accuracy above 98 %, demonstrating both computational efficiency and robustness of the proposed multi-microgrid trading mechanism under large-scale deployment scenarios.

As shown in Fig. 11, the following trends are observed:

- Computation time increases linearly with the number of microgrids, growing from 6.25ms (10 MGs) to 75.41ms (100 MGs), indicating efficient scalability.

Table 11
Scalability analysis of multi-microgrid energy trading.

Microgrid scale	Runtime (ms)	Transaction throughput (T/ms)	Energy matching accuracy (%)
10 MGs	6.25	1.44	99.43
50 MGs	32.98	1.30	98.38
100 MGs	75.41	1.21	98.10

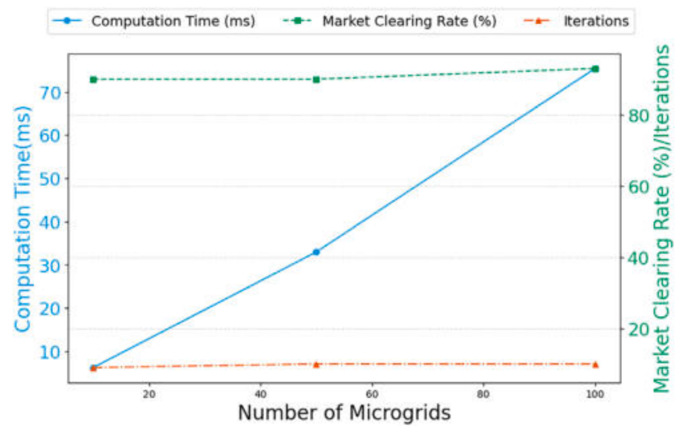

Fig. 11. Scalability analysis of multi-microgrid energy trading.

- Market clearing rate remains stable at around 90 % for small and medium markets and slightly increases to 93 % in large markets, demonstrating the robustness of the matching process.
- Iterations required for convergence remain consistent (9–10 iterations), suggesting that the mechanism maintains efficiency even as the market size expands.

These results confirm that the proposed trading mechanism effectively scales to larger multi-microgrid systems while maintaining computational efficiency and high market-clearing performance.

8. Conclusion

This paper proposes a novel multi-microgrid energy trading mechanism based on cross-chain technology and reputation-based consensus. We designed a matching mechanism based on SDR and MEI to dynamically optimize energy transactions, integrating price adjustment strategies to enhance transaction efficiency. Furthermore, we introduced a reputation-based consensus mechanism to improve fairness and security in decentralized energy trading. A cross-chain-based system was implemented to ensure transaction transparency and prevent manipulation.

Extensive simulations were conducted to evaluate the proposed method. The experimental results demonstrate that:

- The proposed matching mechanism significantly improves social welfare ($WF = 17.48$), outperforming traditional single-round and double auction mechanisms.
- The price adjustment strategy effectively enhances market efficiency, leading to a better match between supply and demand while reducing reliance on the utility grid, and it improves social welfare.
- The reputation-based consensus mechanism mitigates the influence of malicious participants, increasing transaction fairness and security.
- The cross-chain integrated system provides a secure and transparent trading environment, reducing market manipulation risks and increasing overall participant trust.

- The scalability analysis confirms that the proposed mechanism can handle large-scale microgrid networks efficiently, with reasonable computational overhead.

In addition, this work highlights the novelty of the proposed cross-chain architecture. Specifically, the relay chain serves as a decentralized and trustless cross-chain communication and coordination hub, enabling seamless interaction among independently operated microgrid blockchains. Unlike conventional oracle- or sidechain-based interoperability frameworks, our relay chain allows for reputation-weighted consensus participation and smart contract-driven transaction execution, thereby enhancing interoperability, resilience, and system scalability. This design lays the foundation for secure and extensible multi-microgrid energy trading across heterogeneous blockchain environments.

Despite these advantages, this study has some limitations. One notable limitation is the lack of explicit modeling of uncertainty, including the variability of renewable energy sources, load fluctuations, market price volatility, and potential component failures. In this work, we assume a day-ahead market context with a 24-h planning horizon, where renewable generation and demand are based on forecast values, and transaction prices are determined through peer negotiation rather than market clearing prices. While this simplification enables a focused analysis of trading mechanism design, it may limit the robustness and applicability of the model in more dynamic real-world scenarios.

To address this, future work will explore two extensions to enhance uncertainty resilience. First, we plan to introduce a joint energy and reserve trading module using chance-constrained optimization, which accounts for renewable generation uncertainty and guarantees real-time system reliability. Second, we will consider data-driven robust optimization techniques that define uncertainty sets from historical data and aim to find optimal trading strategies under worst-case conditions. These enhancements will improve the resilience, adaptability, and generalizability of the proposed mechanism in uncertain energy environments.

Furthermore, the cross-chain-based system introduces additional transaction costs and latency, which should be carefully evaluated for large-scale deployment. For future work, we also aim to enhance the price adjustment mechanism by incorporating machine learning models for dynamic bidding strategies and integrating real-world market data to validate the framework in practical scenarios.

Overall, this study provides a promising solution for decentralized energy trading, contributing to the development of sustainable and efficient multi-microgrid markets.

CRediT authorship contribution statement

Jiajian Zhu: Writing – original draft. **Zhenyu Guan:** Validation. **Haibin Zheng:** Validation. **Zhenwei Guo:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proof of convergence

This section presents the theoretical proof of convergence for the proposed price adjustment and preference-based stable matching mechanism within the multi-microgrid P2P energy trading framework. By employing iterative price updates and dynamically adjusted bidirectional stable matching procedures, the algorithm is shown to converge to a market equilibrium in finite steps.

A.1. Assumptions and initial conditions

Consider a basic bilateral energy trading scenario involving a single buyer and a single seller. Let the buyer's initial bid price be $p_b^{(0)}$ and the seller's initial offer price be $p_s^{(0)}$, where $p_b^{(0)} < p_s^{(0)}$, indicating that a trade cannot occur at the initial step.

To facilitate negotiation, the following dynamic price adjustment mechanism is introduced:

$$p_s^{k+1} = p_s^k(1-\mu_k) \tag{65a}$$

$$p_b^{k+1} = p_b^k(1+\mu_k) \tag{65b}$$

Here, $\mu_k > 0$ denotes the price adjustment factor in the k-th round.

A successful match occurs once the buyer's bid meets or exceeds the seller's offer:

$$p_b^{(k)} \geq p_s^{(k)} \tag{66}$$

Define the price gap at iteration k as:

$$\Delta_p^{(k)} = p_s^{(k)} - p_b^{(k)}, \quad \Delta_p^{(0)} > 0 \tag{67}$$

A.2. Iterative dynamics of price gap

From the update rules, the recursive relationship governing the price gap is:

$$\Delta_p^{(k+1)} = p_s^{(k+1)} - p_b^{(k+1)} \tag{68a}$$

$$= p_s^{(k)}(1-\mu_k) - p_b^{(k)}(1+\mu_k) \tag{68b}$$

$$= \Delta_p^{(k)} - \mu_k(p_s^{(k)} + p_b^{(k)}) \tag{68c}$$

Since $\mu_k > 0$ and $p_s^{(k)}, p_b^{(k)} > 0$, we have the price gap:

$$\Delta_p^{(k+1)} < \Delta_p^{(k)} \tag{69}$$

Thus, the price gap strictly decreases with each iteration.

A.3. Asymptotic convergence

The difference in price gap per iteration is given by:

$$\Delta_p^{(k)} - \Delta_p^{(k+1)} = \mu_k(p_s^{(k)} + p_b^{(k)}) \tag{70}$$

As each decrement is strictly positive and the sequence is bounded below by zero, the price gap constitutes a strictly decreasing and bounded sequence. Therefore, it converges:

$$\lim_{k\to\infty} \Delta_p^{(k)} = 0 \tag{71}$$

Consequently, as the iterations proceed and the condition $p_b^{(k)} \geq p_s^{(k)}$ is eventually met, the buyer and seller converge to a consensus price, indicating that market equilibrium has been achieved.

A.4. Convergence within finite rounds

In practice, to ensure computational feasibility, a tolerance threshold $\epsilon > 0$ is defined. Once $\Delta_p^{(k)} \leq \epsilon$, the mechanism is considered to have converged. From Eqn. (68c), we derive the expression:

$$\Delta_p^{(k)} = \Delta_p^{(0)} \prod_{r=0}^{k} \left(1 - \mu_r \frac{p_s^{(r)} + p_b^{(r)}}{\Delta_p^{(r)}}\right) \tag{72}$$

Given a sufficiently large adjustment factor μ_r, the price gap $\Delta_p^{(k)}$ decays geometrically. Consequently, for any predefined tolerance $\epsilon > 0$, there exists a finite number of iterations N such that:

$$\Delta_p^{(N)} \leq \epsilon \tag{73}$$

As a result, the proposed mechanism ensures that the market reaches near-equilibrium in a finite and computationally efficient manner.

A.5. Convergence of the matching algorithm

Beyond price adjustment, the overall matching algorithm ensures convergence by progressively eliminating unstable matches. In each round, every buyer proposes to sellers in their preference list. Sellers retain only the most preferred offers, and reject others. This stable matching procedure eliminates infeasible matches over iterations.

Because the number of microgrids and their preference lists is finite, each microgrid is limited to a finite number of potential partners, and the process is guaranteed to terminate in finite number of rounds. The final outcome constitutes a stable matching satisfying individual rationality and blocking pair conditions. Thus, the matching mechanism is also guaranteed to converge.

A.6. Completeness and fallback mechanism

To ensure the robustness and completeness of the trading framework, any transactions that remain unresolved after multiple rounds of P2P negotiation are redirected to the utility grid for settlement. This fallback mechanism ensures that no energy supply or demand is left unaddressed, thereby maintaining overall system balance and guaranteeing the finality of all transactions.

In conclusion, the proposed pricing and matching mechanisms are theoretically sound and practically effective, ensuring convergence and achieving a robust, efficient, and fair market equilibrium in decentralized multi-microgrid energy trading scenarios.

Data availability

No data were used for the research described in the article.

References

[1] Boyd S, Parikh N, Chu E, Peleato B, Eckstein J, et al. Distributed optimization and statistical learning via the alternating direction method of multipliers. Found. Trends Mach. Learn. 2011;3(1):1–122.
[2] Han B, Zahraoui Y, Mubin M, Mekhilef S, Korõtko T, Alshammari O. Distributed optimal storage strategy in the ADMM-based peer-to-peer energy trading considering degradation cost. J Energy Storage 2024;96:112651.
[3] Samende C, Cao J, Fan Z. Multi-agent deep deterministic policy gradient algorithm for peer-to-peer energy trading considering distribution network constraints. Appl Energy 2022;317:119123.
[4] Wei E, Ozdaglar A. Distributed alternating direction method of multipliers. In: 2012 IEEE 51st IEEE conference on decision and control (CDC); IEEE; 2012. p. 5445–50.
[5] Ananduta W, Ocampo-Martinez C, Nedić A. A distributed augmented lagrangian method over stochastic networks for economic dispatch of large-scale energy systems. IEEE Trans Sustain Energy 2021;12(4):1927–34.
[6] Xu Y, Sun H, Gu W. A novel discounted min-consensus algorithm for optimal electrical power trading in grid-connected dc microgrids. IEEE Trans Ind Electron 2019;66(11):8474–84.
[7] AlSkaif T, Crespo-Vazquez JL, Sekuloski M, Van Leeuwen G, Catalão JP. Blockchain-based fully peer-to-peer energy trading strategies for residential energy systems. IEEE Trans Ind Informatics 2021;18(1):231–41.
[8] Nwulu NI, Xia X. Optimal dispatch for a microgrid incorporating renewables and demand response. Renew Energy 2017;101:16–28.
[9] Santos LHS, Silva JAA, López JC, Arias NB, Rider MJ, Da Silva LCP. Integrated optimal sizing and dispatch strategy for microgrids using homer pro. In: 2021 IEEE PES innovative smart grid technologies conference-latin america (ISGT latin america); IEEE; 2021. p. 1–5.
[10] Khorasany M, Mishra Y, Ledwich G. A decentralized bilateral energy trading system for peer-to-peer electricity markets. IEEE Trans Ind Electron 2019;67(6):4646–57.
[11] Sorin E, Bobo L, Pinson P. Consensus-based approach to peer-to-peer electricity markets with product differentiation. IEEE Trans Power Syst 2018;34(2):994–1004.
[12] Baroche T, Pinson P, Latimier RLG, Ahmed HB. Exogenous cost allocation in peer-to-peer electricity markets. IEEE Trans Power Syst 2019;34(4):2553–64.
[13] Guo Z, Pinson P, Chen S, Yang Q, Yang Z. Chance-constrained peer-to-peer joint energy and reserve market considering renewable generation uncertainty. IEEE Trans Smart Grid 2020;12(1):798–809.
[14] Guo Z, Pinson P, Chen S, Yang Q, Yang Z. Online optimization for real-time peer-to-peer electricity market mechanisms. IEEE Trans Smart Grid 2021;12(5):4151–63.
[15] Guo Z, Pinson P, Wu Q, Chen S, Yang Q, Yang Z. An asynchronous online negotiation mechanism for real-time peer-to-peer electricity markets. IEEE Trans Power Syst 2021;37(3):1868–80.
[16] Masaud TM, Warner J, El-Saadany EF. A blockchain-enabled decentralized energy trading mechanism for islanded networked microgrids. IEEE Access 2020;8:211291–302.
[17] Wei C, Shen Z, Xiao D, Wang L, Bai X, Chen H. An optimal scheduling strategy for peer-to-peer trading in interconnected microgrids based on ro and nash bargaining. Appl Energy 2021;295:117024.
[18] Wei C, Wang Y, Shen Z, Xiao D, Bai X, Chen H. Auq–ADMM algorithm-based peer-to-peer trading strategy in large-scale interconnected microgrid systems considering carbon trading. IEEE Syst J 2023;17(4):6248–59.
[19] Lee J, Guo J, Choi JK, Zukerman M. Distributed energy trading in microgrids: a game-theoretic model and its equilibrium analysis. IEEE Trans Ind Electron 2015;62(6):3524–33.
[20] Jadhav AM, Patne NR, Guerrero JM. A novel approach to neighborhood fair energy trading in a distribution network of multiple microgrid clusters. IEEE Trans Ind Electron 2018;66(2):1520–31.
[21] Andoni M, Robu V, Flynn D, Abram S, Geach D, Jenkins D, et al. Blockchain technology in the energy sector: a systematic review of challenges and opportunities. Renew Sustain Energy Rev 2019;100:143–74.
[22] Ping J, Yan Z, Chen S. A privacy-preserving blockchain-based method to optimize energy trading. IEEE Trans Smart Grid 2022;14(2):1148–57.
[23] Umar A, Kumar D, Ghose T. Blockchain-based decentralized energy intra-trading with battery storage flexibility in a community microgrid system. Appl Energy 2022;322:119544.
[24] Zhong X, Liu Y, Xie K, Xie S. A local electricity and carbon trading method for multi-energy microgrids considering cross-chain interaction. Sensors 2022;22(18):6935.
[25] Zhang S, Hou C. Model of decentralized cross-chain energy trading for power systems. Glob Energy Interconnect 2021;4(3):324–34.
[26] Wang Y, Xie H, Sun X, Tang L, Bie Z. A cross-chain enabled day-ahead collaborative power-carbon-TGC market. Energy 2022;258:124881.
[27] Ghaddar B, Ljubić I, Qiu Y. Three network design problems for community energy storage. Networks 2024;84(4):420–45.
[28] Daneshvar M, Mohammadi-Ivatloo B, Zare K. An innovative transactive energy architecture for community microgrids in modern multi-carrier energy networks: a Chicago case study. Sci Rep 2023;13(1):1529.
[29] Farivar M, Low SH. Branch flow model: relaxations and convexification—part i. IEEE Trans Power Syst 2013;28(3):2554–64.
[30] Farivar M, Low SH. Branch flow model: relaxations and convexification—part ii. IEEE Trans Power Syst 2013;28(3):2565–72.
[31] Saadat H, et al. Power system analysis. Vol. 2. McGraw-hill; 1999.
[32] Zhang X, Ge S, Liu H, Zhou Y, He X, Xu Z. Distributionally robust optimization for peer-to-peer energy trading considering data-driven ambiguity sets. Appl Energy 2023;331:120436.
[33] Mansouri SA, Nematbakhsh E, Ramos A, Tostado-Véliz M, Aguado JA, Aghaei J. A robust ADMM-enabled optimization framework for decentralized coordination of microgrids. IEEE Trans Ind Informatics 2024.
[34] Wang J, Zheng J, Yu L, Goh M, Tang Y, Huang Y. Distributed reputation-distance iterative auction system for peer-to-peer power trading. Appl Energy 2023;345:121306.